



Or: why are the Dutch so tall?!

# Body shape and size diversity

Dr. Laura van Holstein

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Anthropology 215

# Important reminders

1. Date & time of final exam has been confirmed: **12/11/2025; 8-10.30AM**
2. Midterm exam: **10/16/2025 during class**
3. Revision session for midterm exam: **next week Thursday**
4. Assignment coming up! Due: **10/10/2025 11.59PM**

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  - Choice between Adaptation Matrix OR Red List
  - All information is in syllabus and on Canvas

The screenshot shows the Canvas LMS interface for the course FA25\_ANT\_210\_1. The left sidebar contains navigation links: Home, Announcements, Modules, Assignments (selected), Syllabus, Grades, Namecoach, Studio, Item Banks, Multi-Tool, Course Analytics, Gradescope, Outcomes, Collaborations, and Rubrics. The main content area displays the assignment 'Adaptation Matrix OR Red List' with a 'Published' status and buttons for 'Assign To', 'Edit', and a menu icon. The instructions section lists six steps for completing the assignment, including choosing between the Adaptation Matrix or Red List, downloading forms, and submitting as a PDF by the deadline of Friday 10 October at 11.59PM. Below the instructions, a table shows the assignment details:

| Points     | 5             |                |                   |
|------------|---------------|----------------|-------------------|
| Submitting | a file upload |                |                   |
| Due        | For           | Available from | Until             |
| Oct 10     | Everyone      | -              | Oct 11 at 12:30am |

At the bottom, there is a '+ Rubric' button.

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- **Choice between Adaptation Matrix OR Red List**
- **All information is in syllabus and on Canvas**

*Red list - database of species assessed by  
how likely they are to go extinct*

EMORY  
FA25\_ANT\_210\_1 > Assignments > Adaptation Matrix OR Red List

Fall 2025

Home  
Announcements  
Modules  
Assignments  
Syllabus  
Grades  
NameCoach  
Studio  
Item Banks  
Multi-Tool  
Course Analytics  
Gradescope  
Outcomes  
Collaborations  
Rubrics

### Adaptation Matrix OR Red List

Published Assign To Edit

**INSTRUCTIONS:**

1. Choose which assignment you'd like to complete: the Adaptation Matrix or the Red List.
2. Read the assignment instructions for your chosen assignment which can be found in the most up-to-date syllabus, carefully.
3. Download fillable Adaptation Matrix [here](#), and the fillable Red List form [here](#).
4. Fill in the appropriate document for your chosen assignment according to the instructions.
5. Upload your work as PDF, with the file name specified on the Adaptation Matrix or Red List document, here on Canvas **before the due date and time (Friday 10 October at 11.59PM)**.
6. Late submissions are **not permitted**, unless previous arrangements have been made directly with Dr. Van Holstein at least 48 hours before the deadline.

Points 5  
Submitting a file upload

| Due    | For      | Available from | Until             |
|--------|----------|----------------|-------------------|
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+ Rubric

*assessment  
based on  
ancestral  
pop size,  
life history  
(first reproduction)  
state of environment*



# Thank you Jane

Humans  
are just  
animals part  
of ecosystem



1934-2025





# Body shape and diversity

Dr. Laura van Holstein

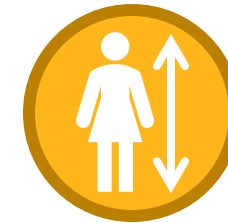
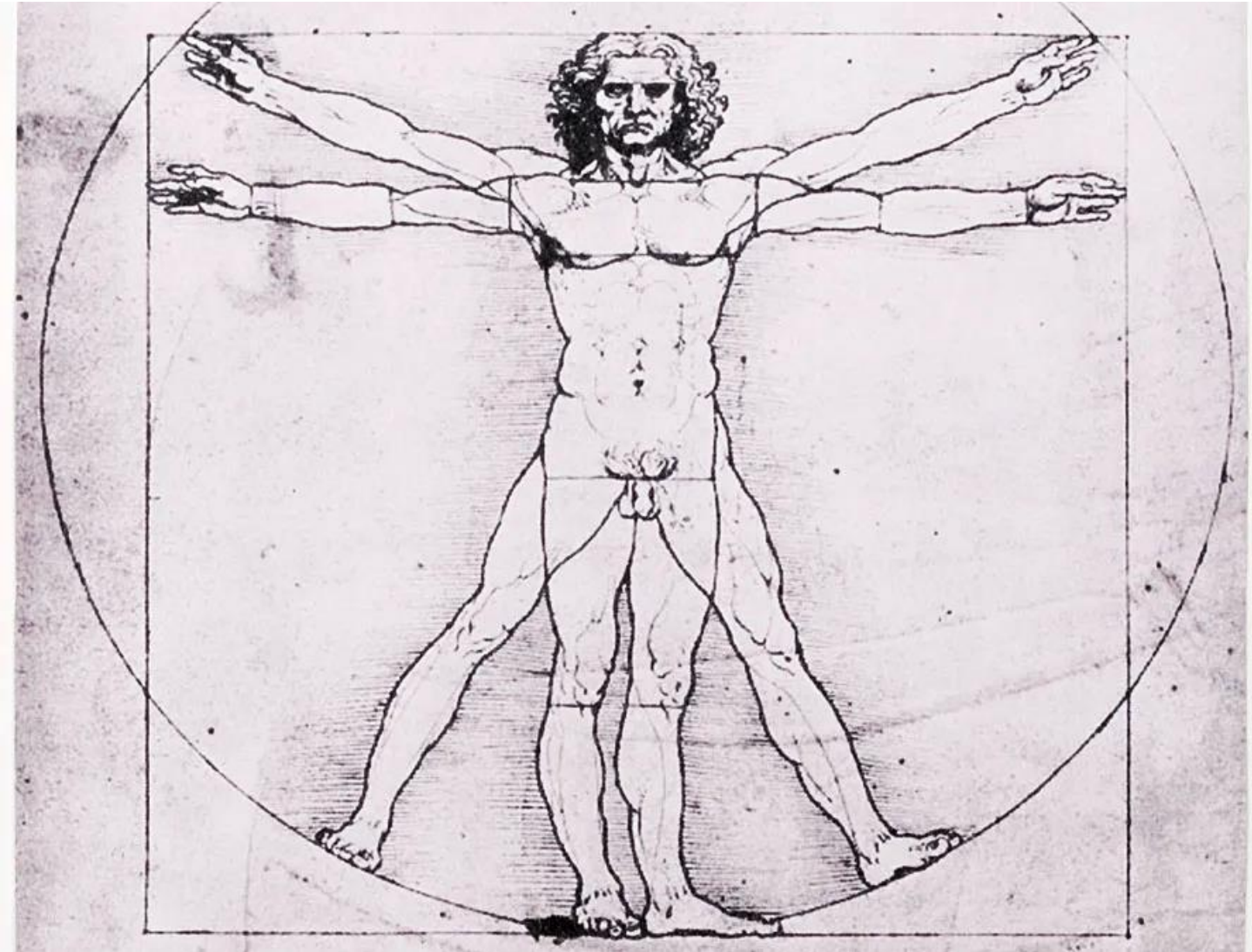
[lavanho@emory.edu](mailto:lavanho@emory.edu)

Anthropology 215

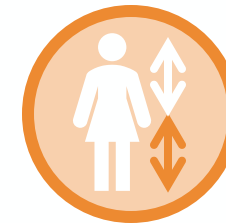




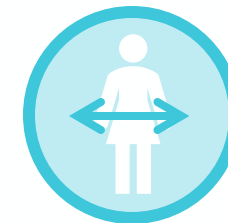
# What can we measure?



**Stature (height)**



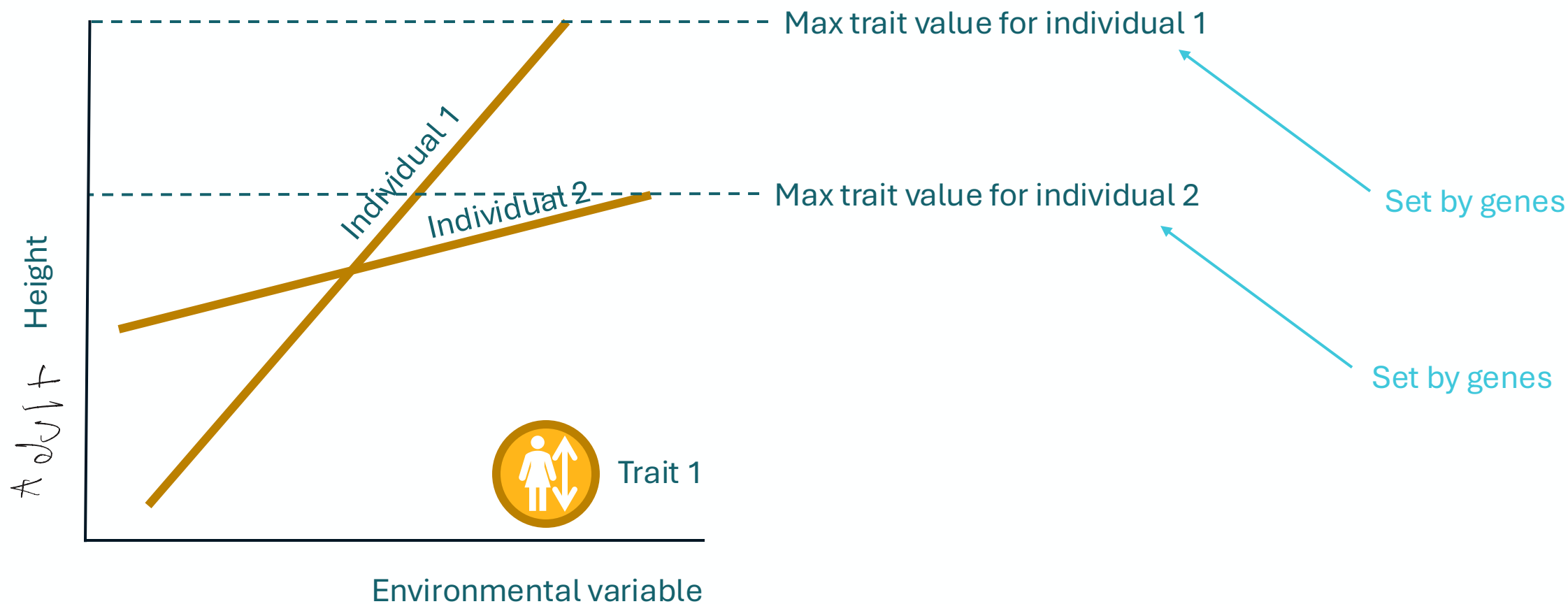
**Proportions**



**Mass**



# Genes set the limits of the phenotype; environment determines whether those limits are reached.



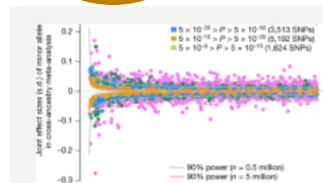


# Genes set the limits of the phenotype; environment determines whether those limits are reached.

Monogenic variants  
clinical cases

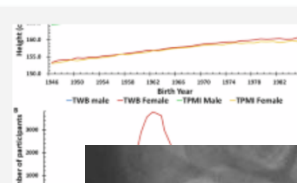


## Stature (height)



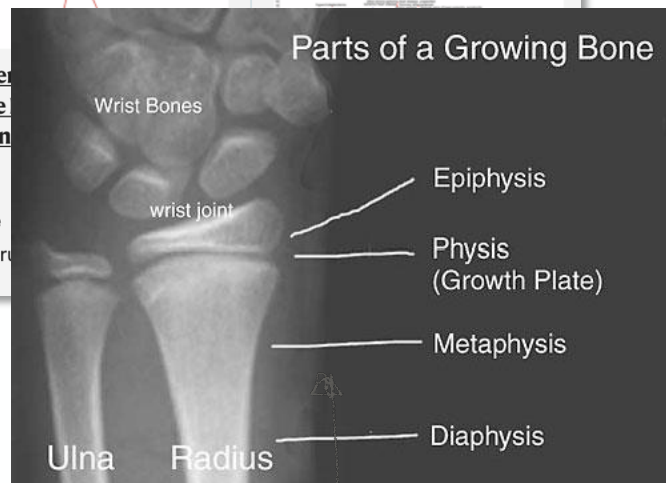
**A saturated map of common genetic variants associated with human height**

Article | Open access  
12 October 2022



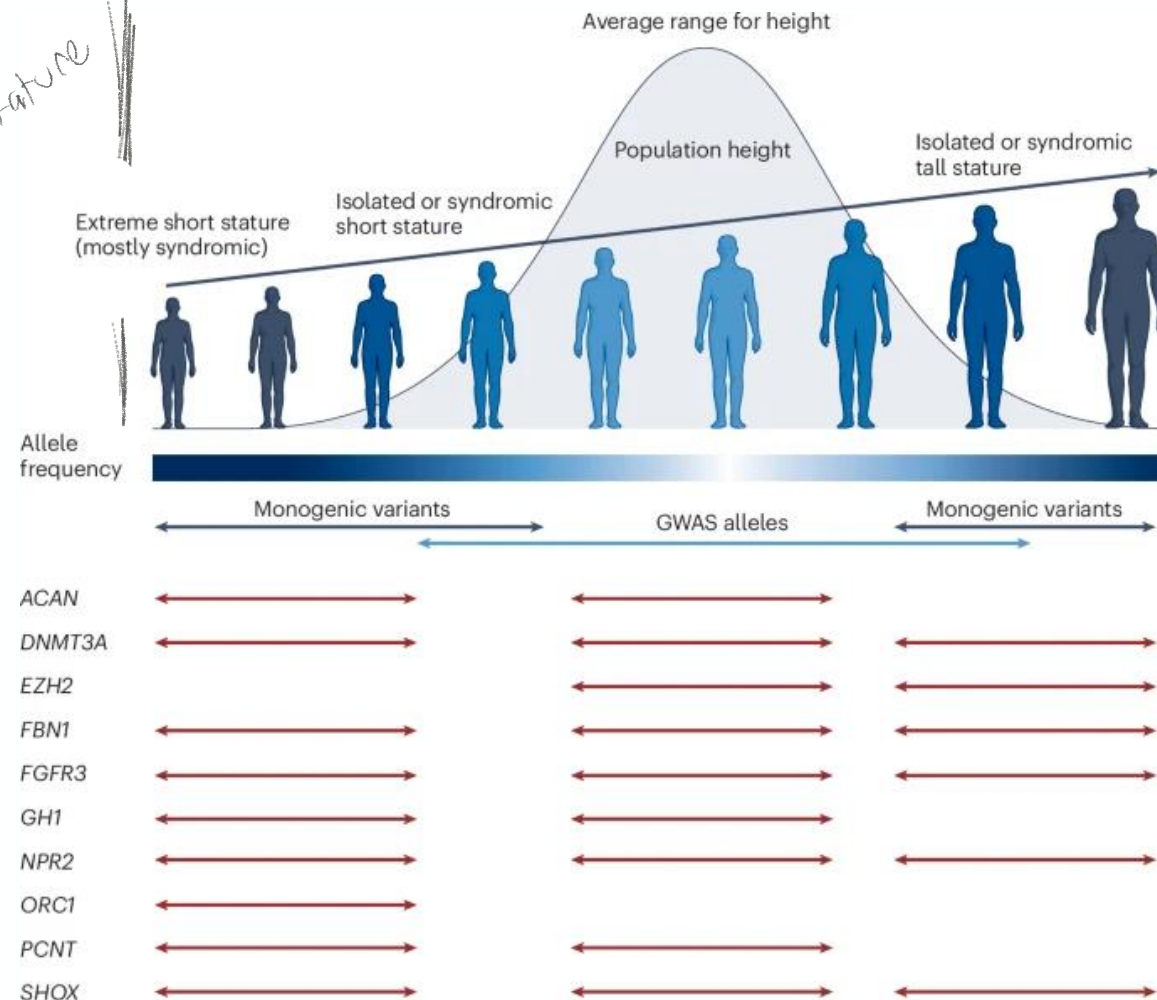
**Polygenic score for the Taiwan population**

Article  
05 February 2023

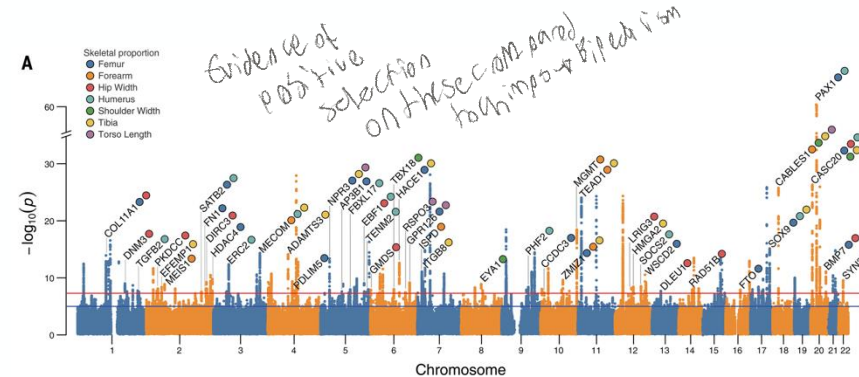


Expression of genes in growth plates for stature

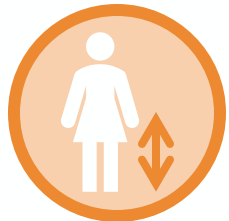
Stature



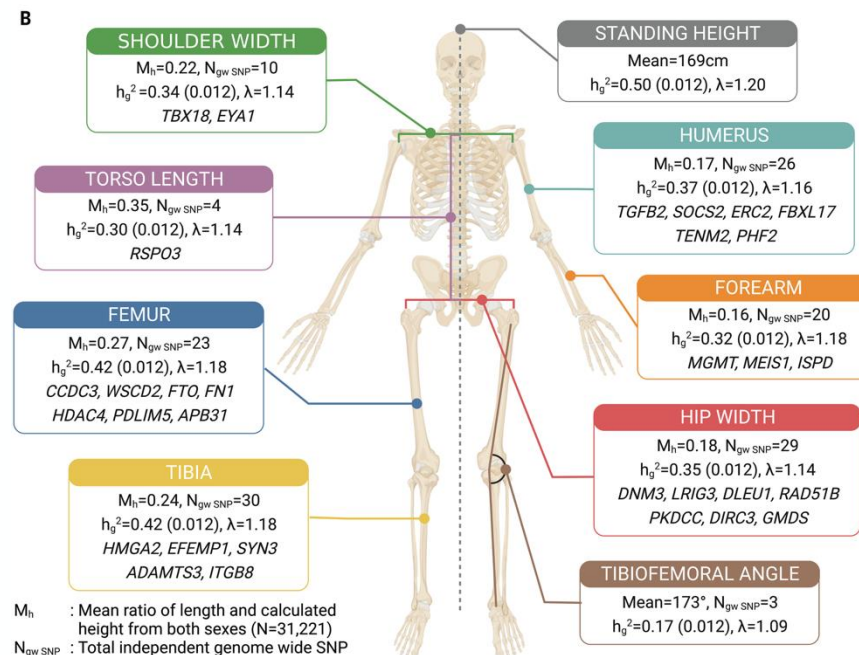
# Genes set the limits of the phenotype; environment determines whether those limits are reached.



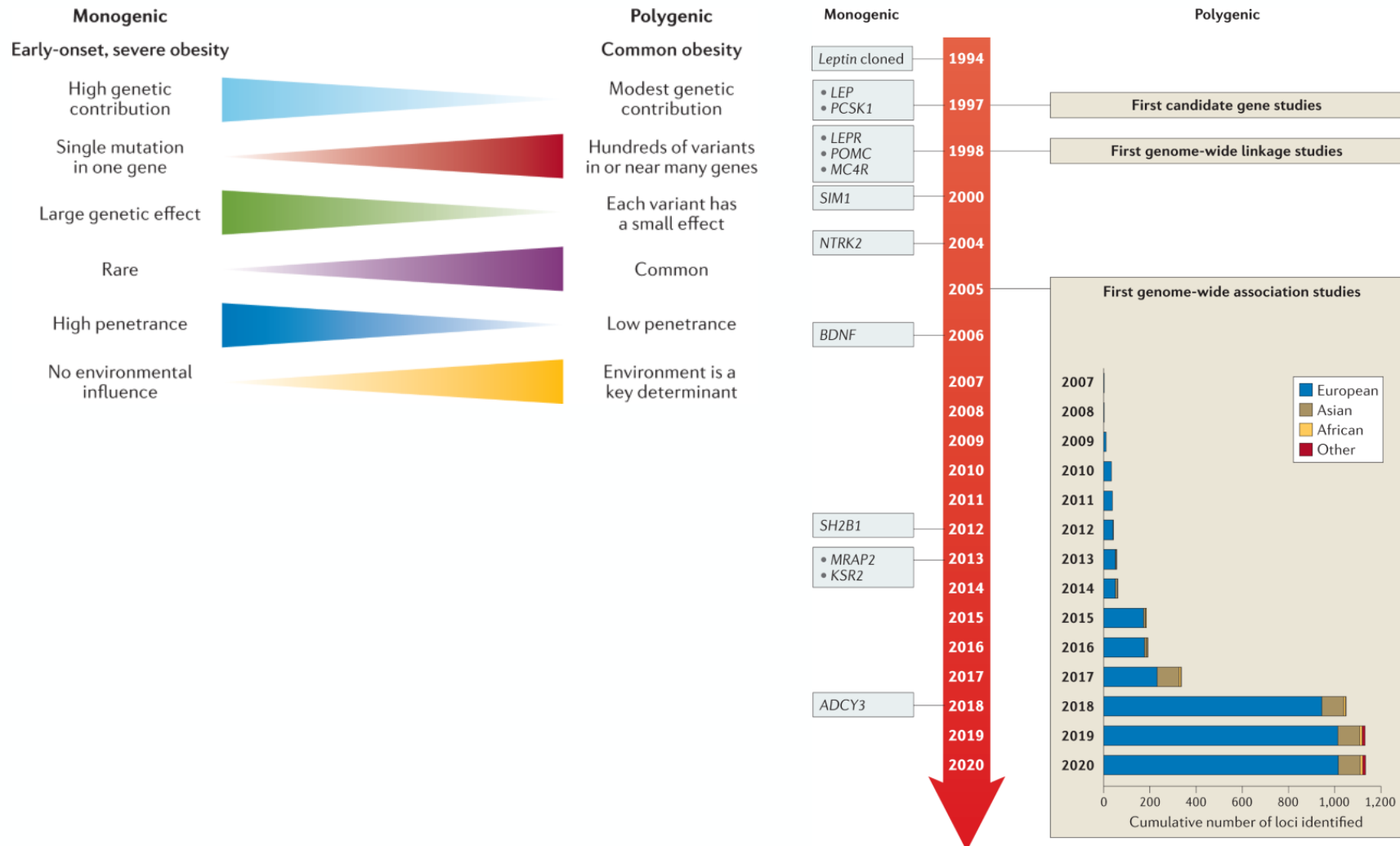
- All skeletal proportions (SPs) are highly heritable (~30 to 50%)
- Genome-wide association studies of these traits identified 145 independent loci
  - Genes that regulate skeletal development as well as those that are associated with rare human skeletal diseases
- In contrast to other traits, loci associated with these SPs are significantly enriched both in human accelerated regions *and* in regulatory elements of genes that are differentially expressed in humans and the great apes



## Proportions



# Genes set the limits of the phenotype; environment determines whether those limits are reached.



Mass



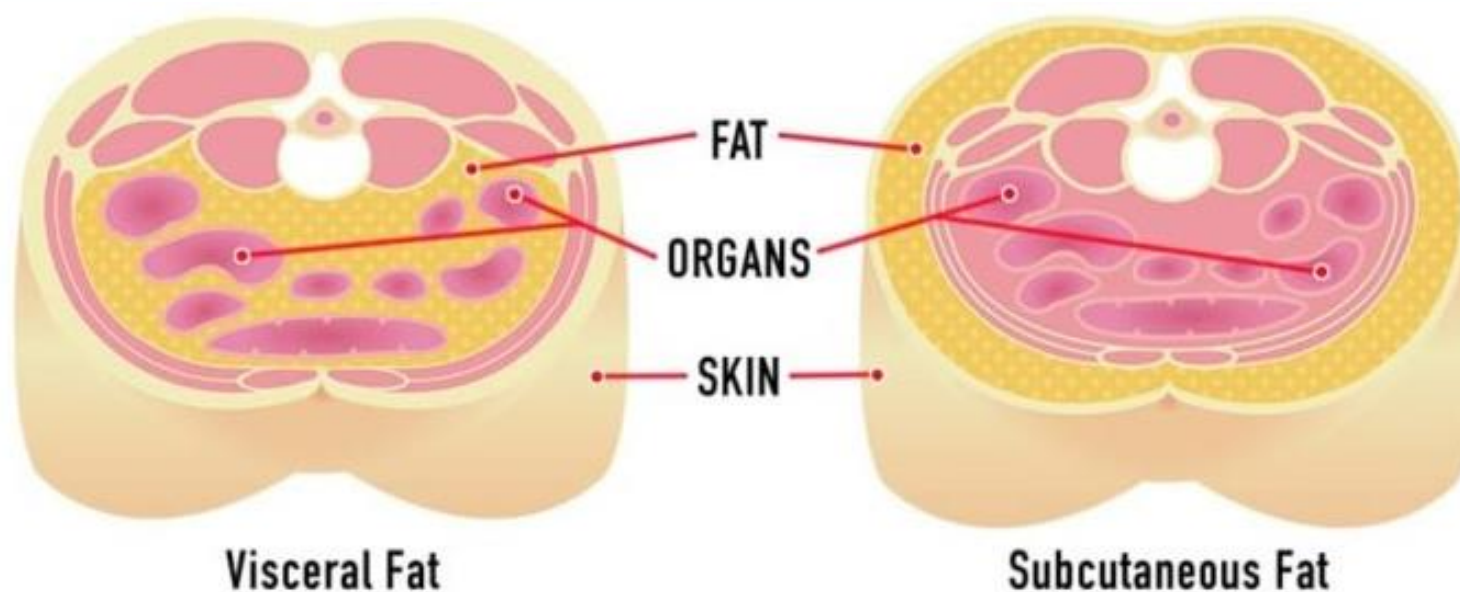
**Genes** set the limits of the phenotype; environment determines whether those limits are reached.

*Predictor of amount of each Fat*

Genes

More androgen receptors:

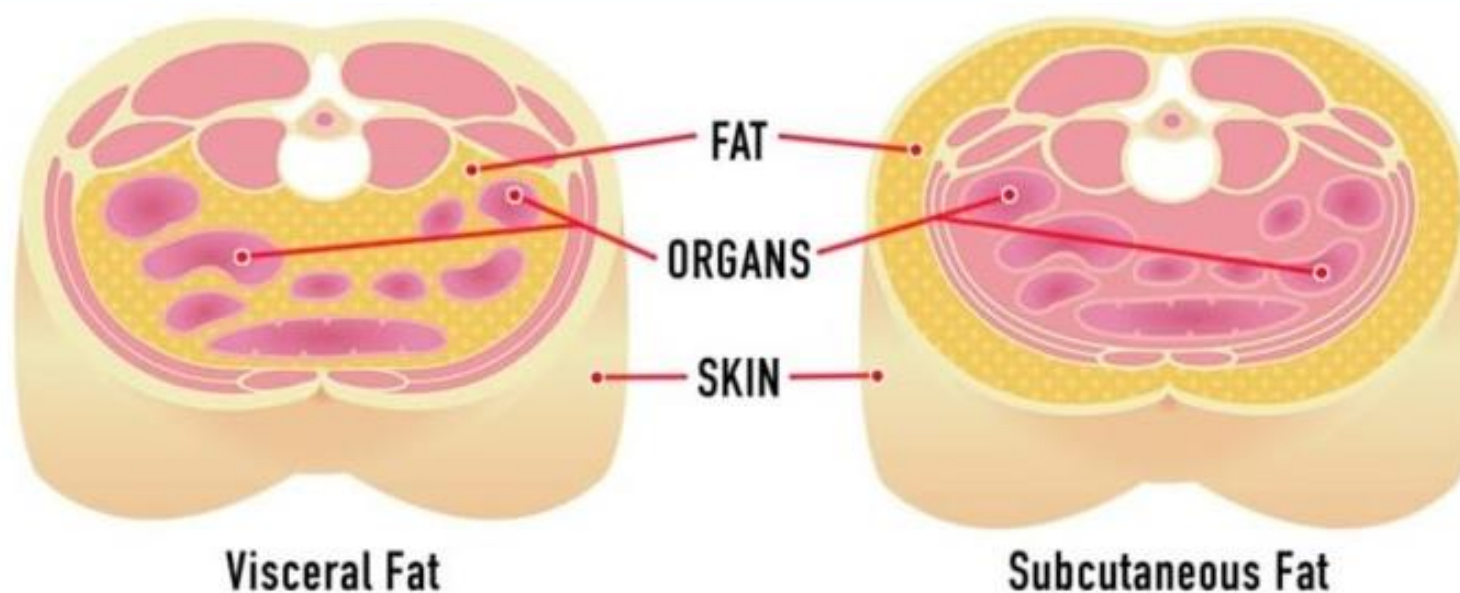
More oestrogen receptors:



Mass

**Genes** set the limits of the phenotype; environment determines whether those limits are reached.

But also... *hormones*  
menopause!  
More androgen receptors:      More oestrogen receptors:

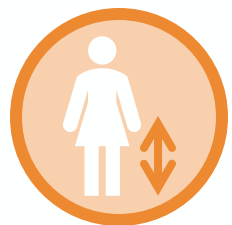


Mass

**Genes** set the limits of the phenotype; environment determines whether those limits are reached.



Stature (height)



Proportions



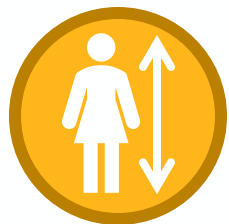
Mass

Somewhat changeable during ontogeny, but set at adulthood

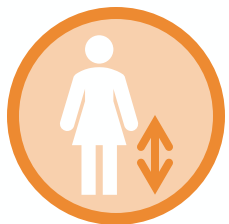
Remains (somewhat) variable throughout life



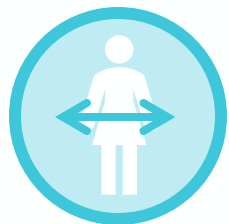
# General patterns in *adult*...



**Stature (height)**



**Proportions**

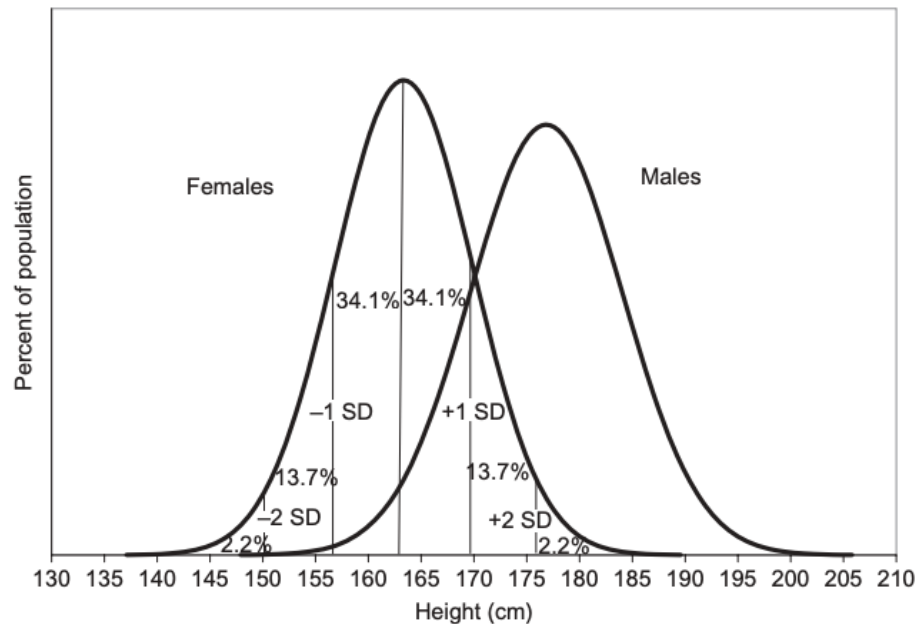


**Mass**

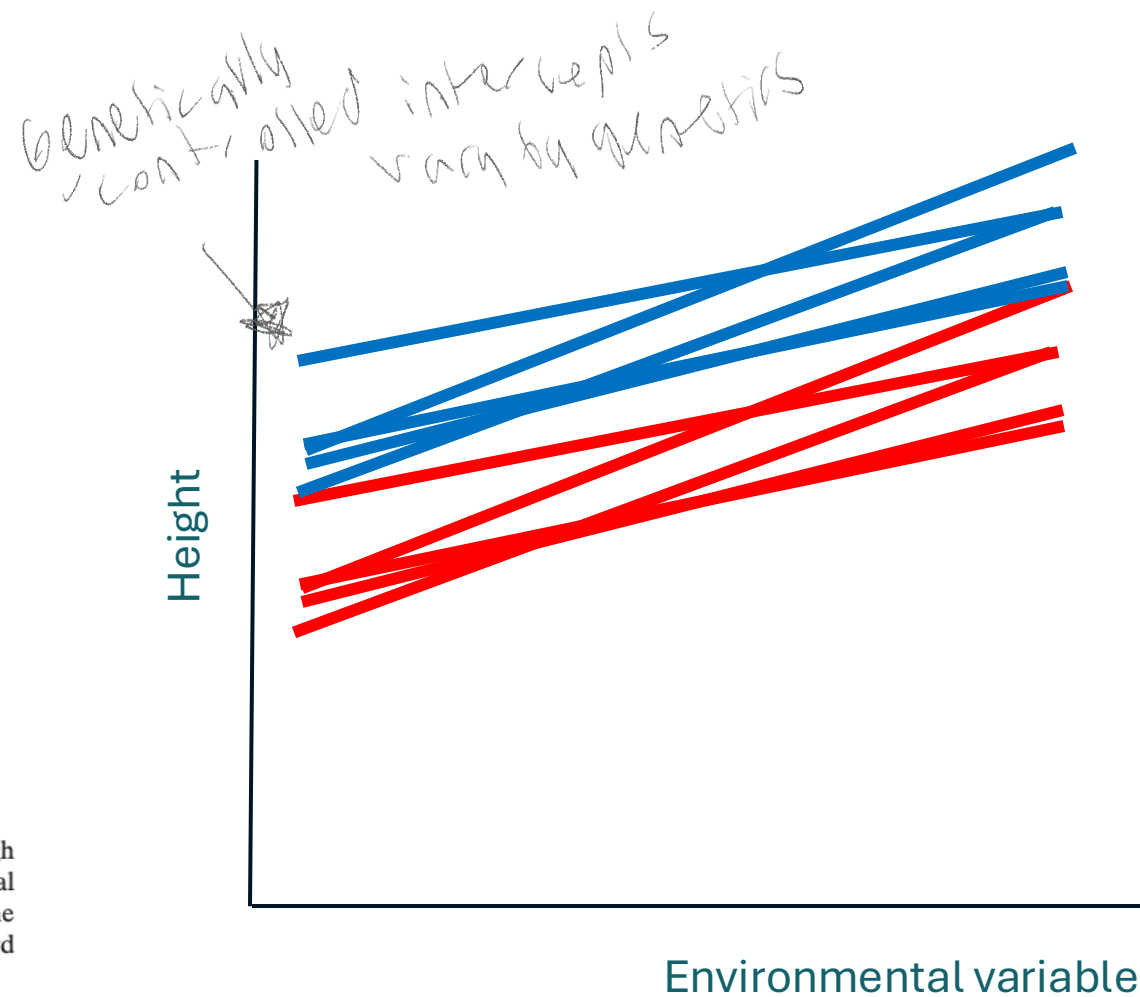


# General patterns

## 1. Males taller than females



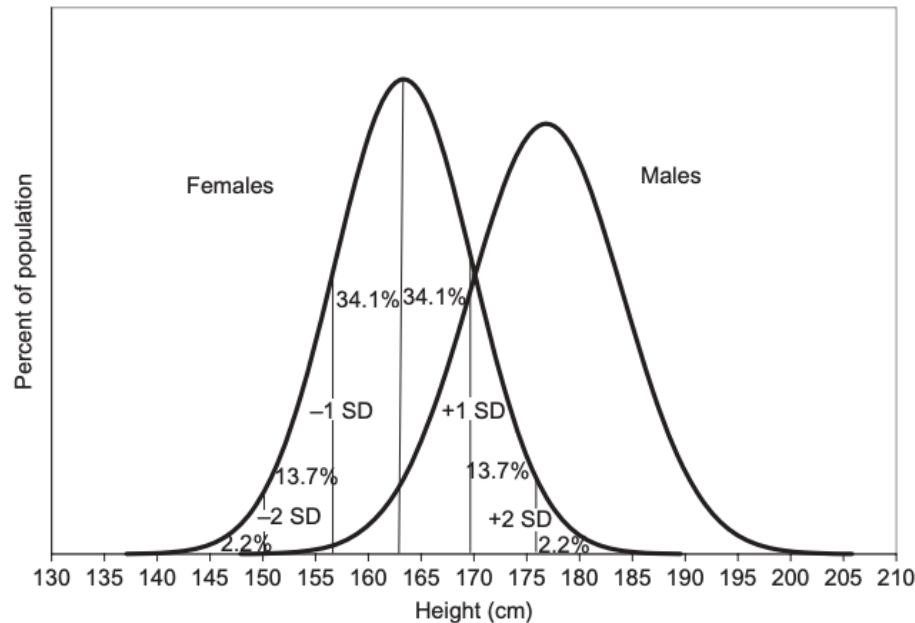
**Figure 12.2** Distribution of adult male and female height in the United States. Although males have a greater mean height than females, distributions overlap considerably. Vertical lines on the female distribution indicate standard deviations (SDs) above and below the mean, and percent values indicate the percentages of population falling between standard deviations (see Box 12.2). *Source:* Data from McDowell et al. (2008).





# General patterns

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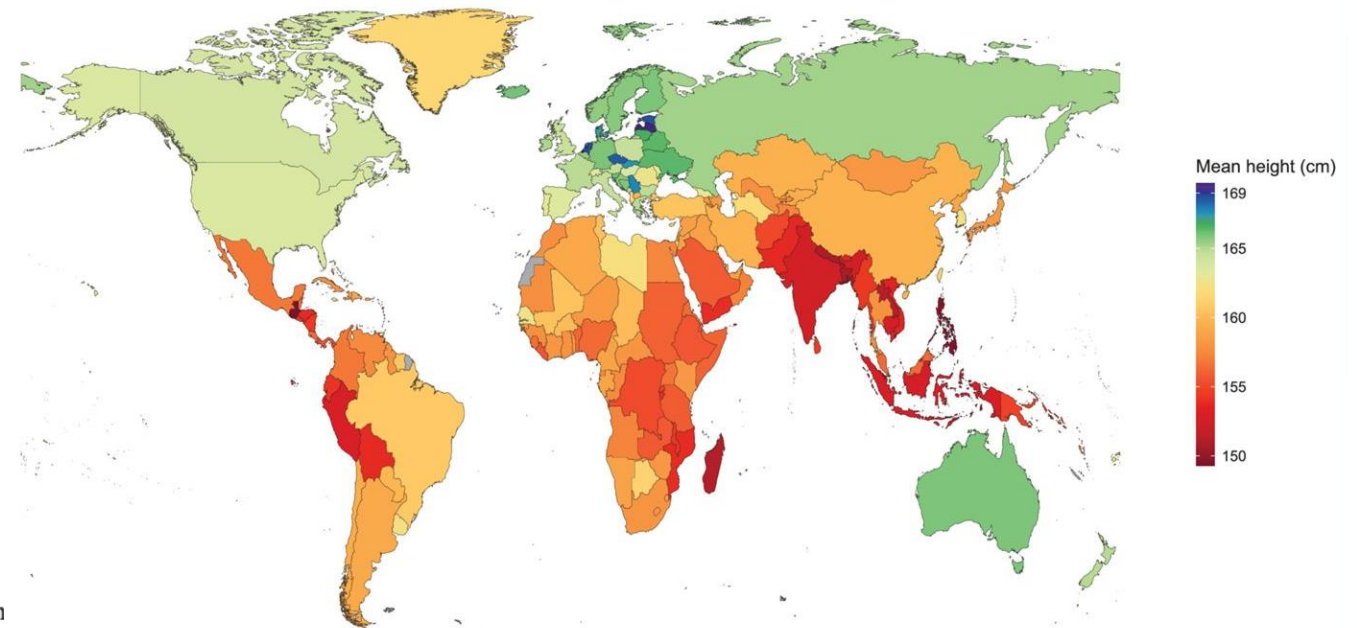


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## 2. a) Differences among populations

### b) Clinal patterns (close together = similar)

1996 birth cohort

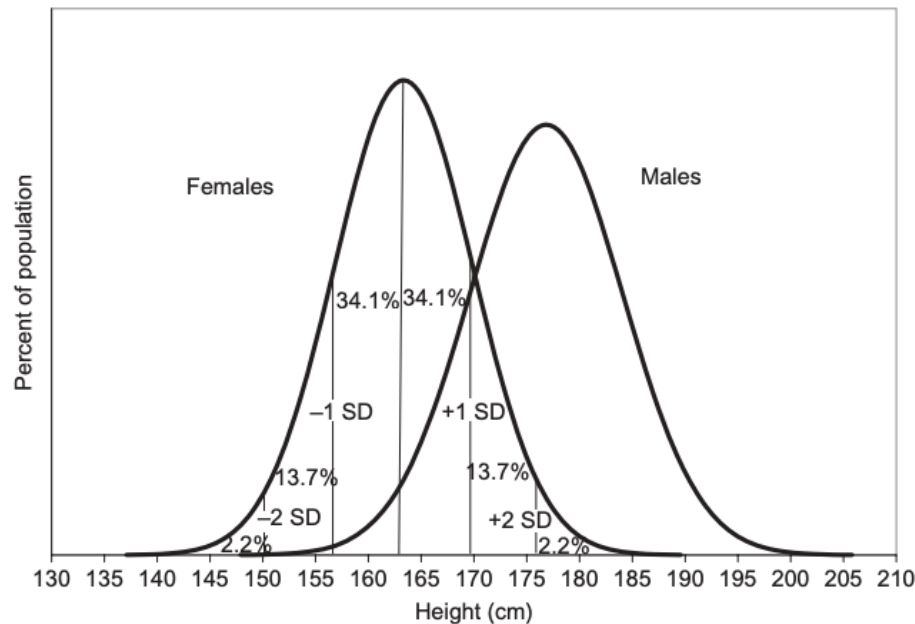






# General patterns

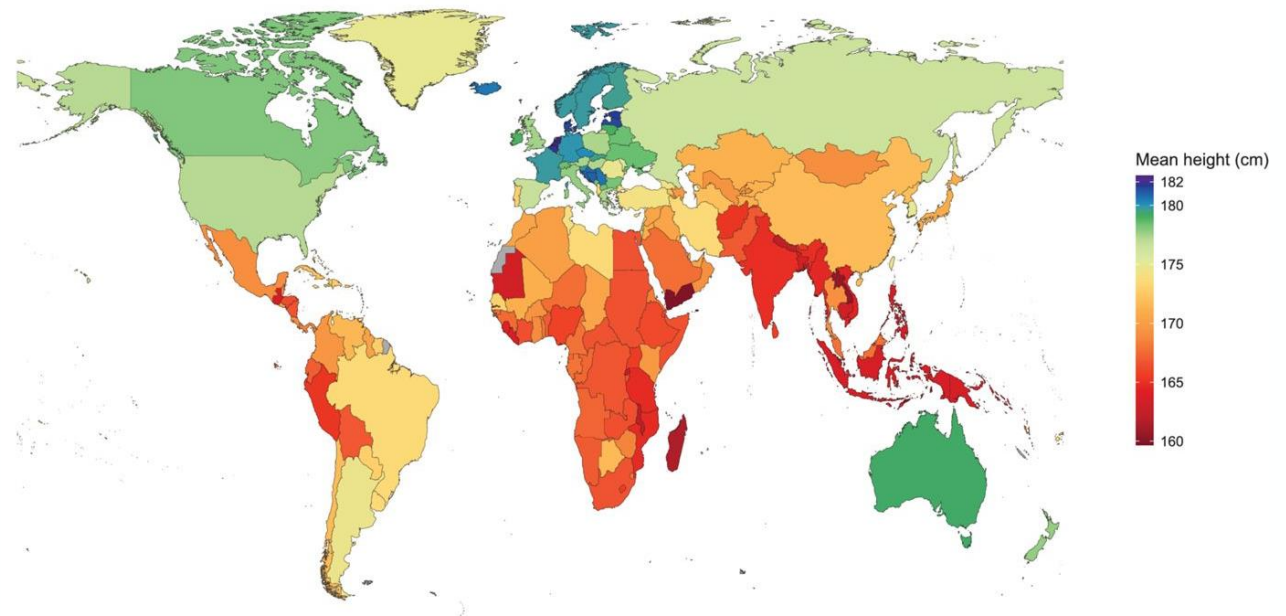
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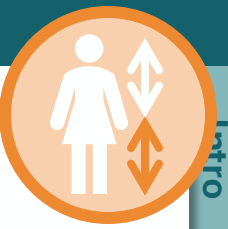


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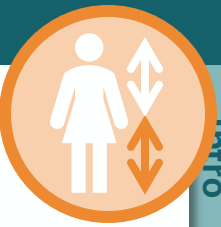




# General patterns

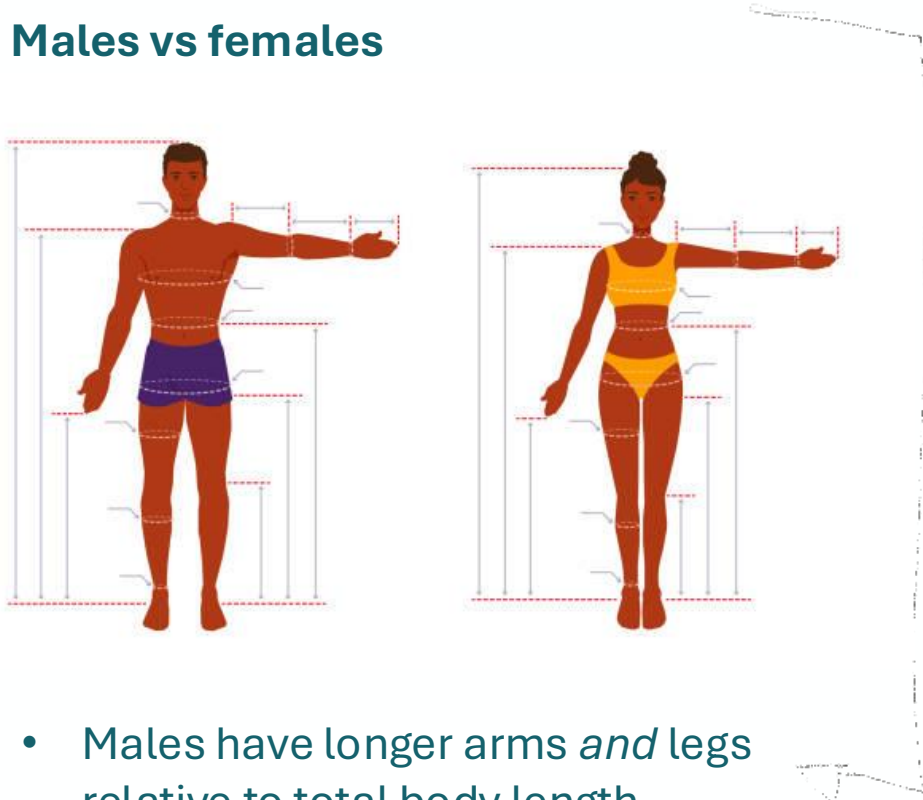
1. Males vs females?

2. a) Differences among populations?  
b) Clinal patterns (close together = similar)?



# General patterns

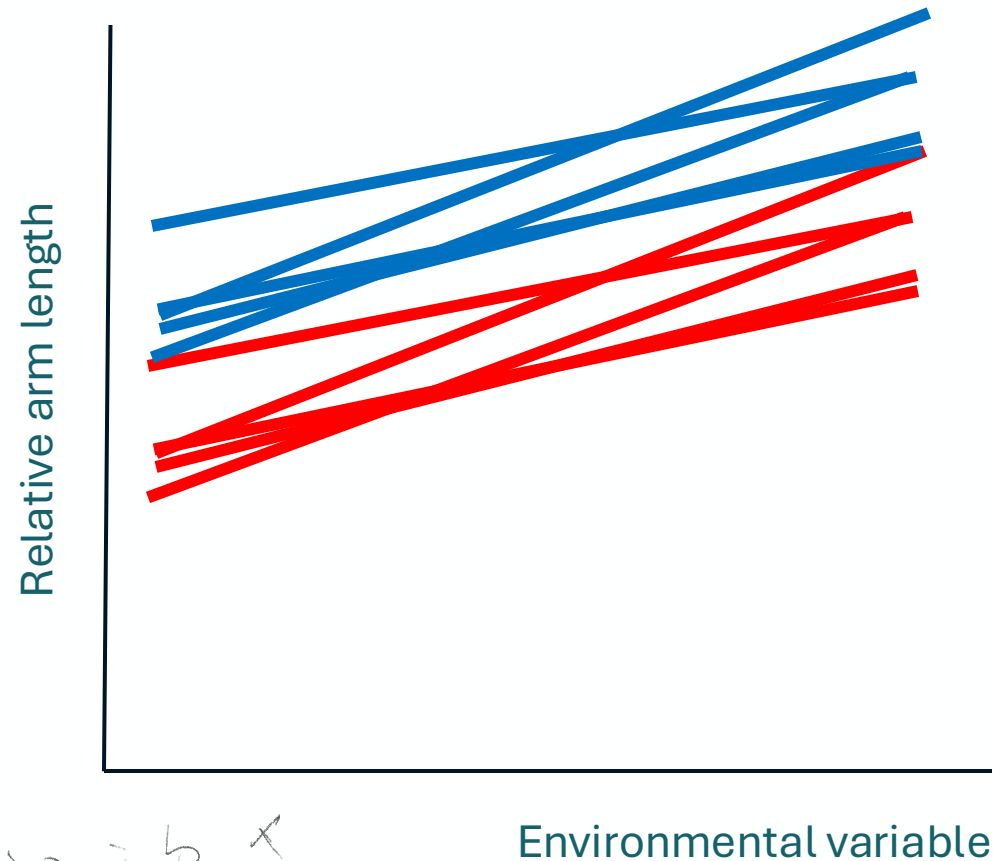
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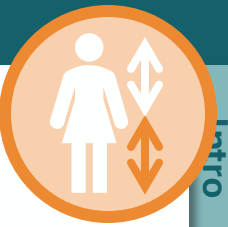


- Males have longer arms *and* legs relative to total body length

still lots of overlap between sexes for limb + stature

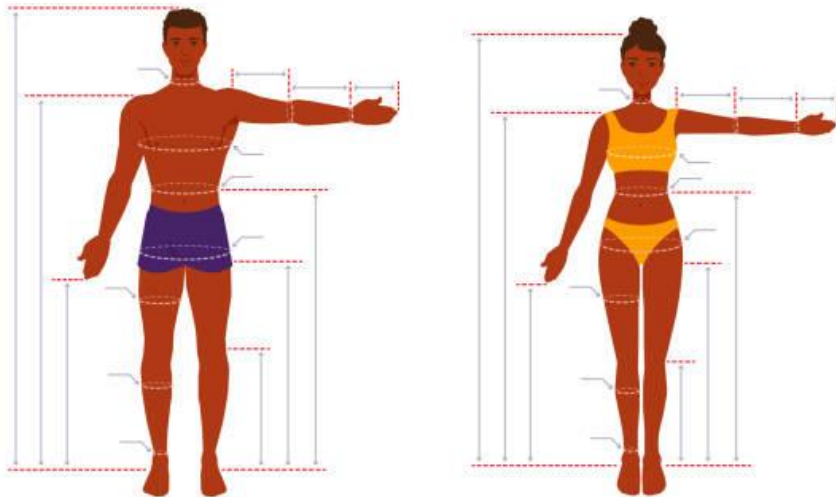
2. a) Differences among populations?  
b) Clinal patterns (close together = similar)?





# General patterns

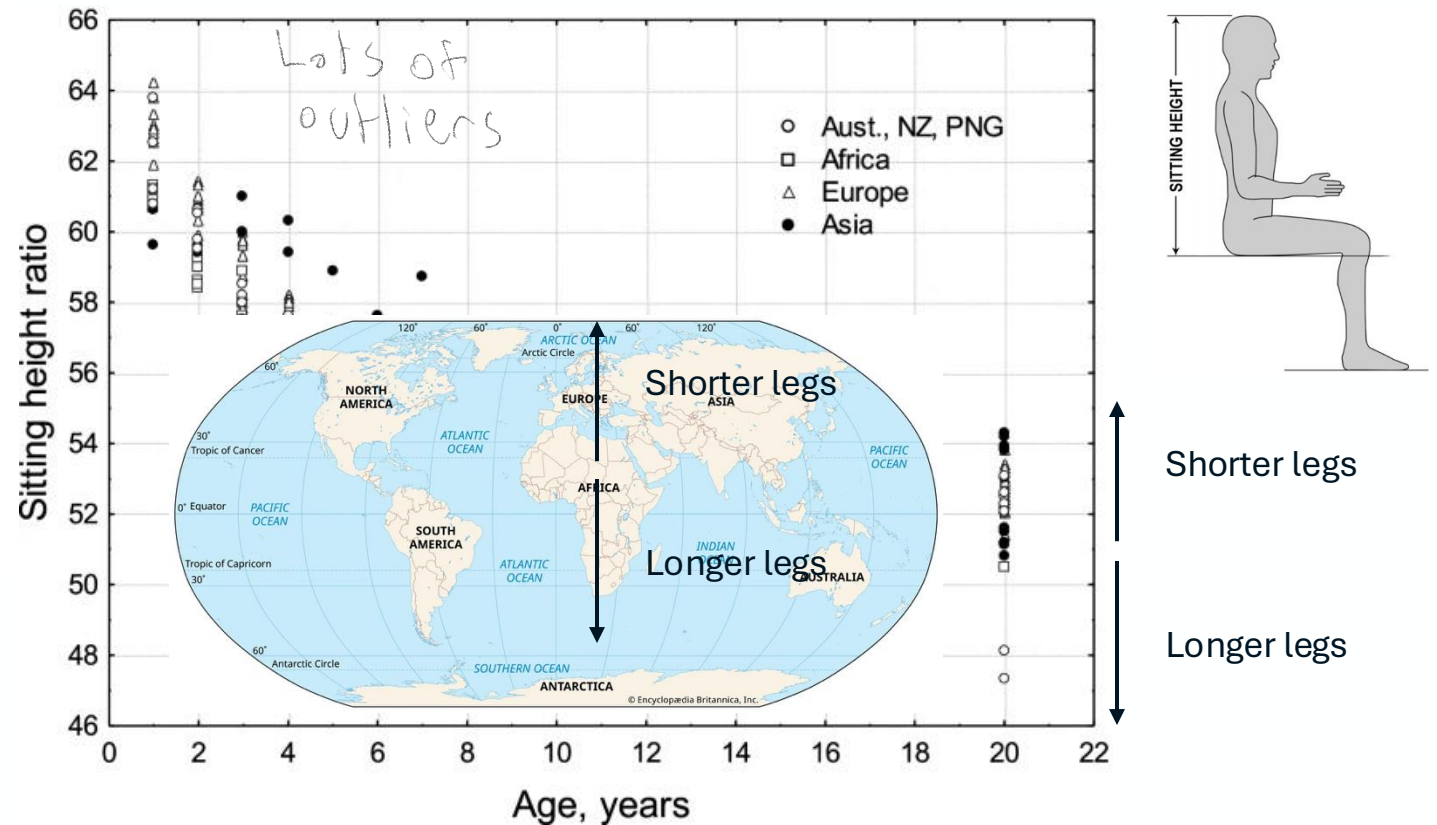
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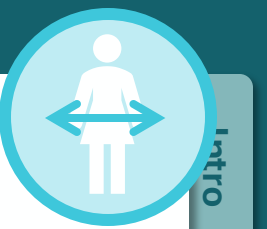
### b) Clinal patterns (close together = similar)



**But: Max SHR (shortest legs): Guatemala Maya & Peruvian women**

**Min SHR (longest legs): Australian First Nations people**





# General patterns

BMI

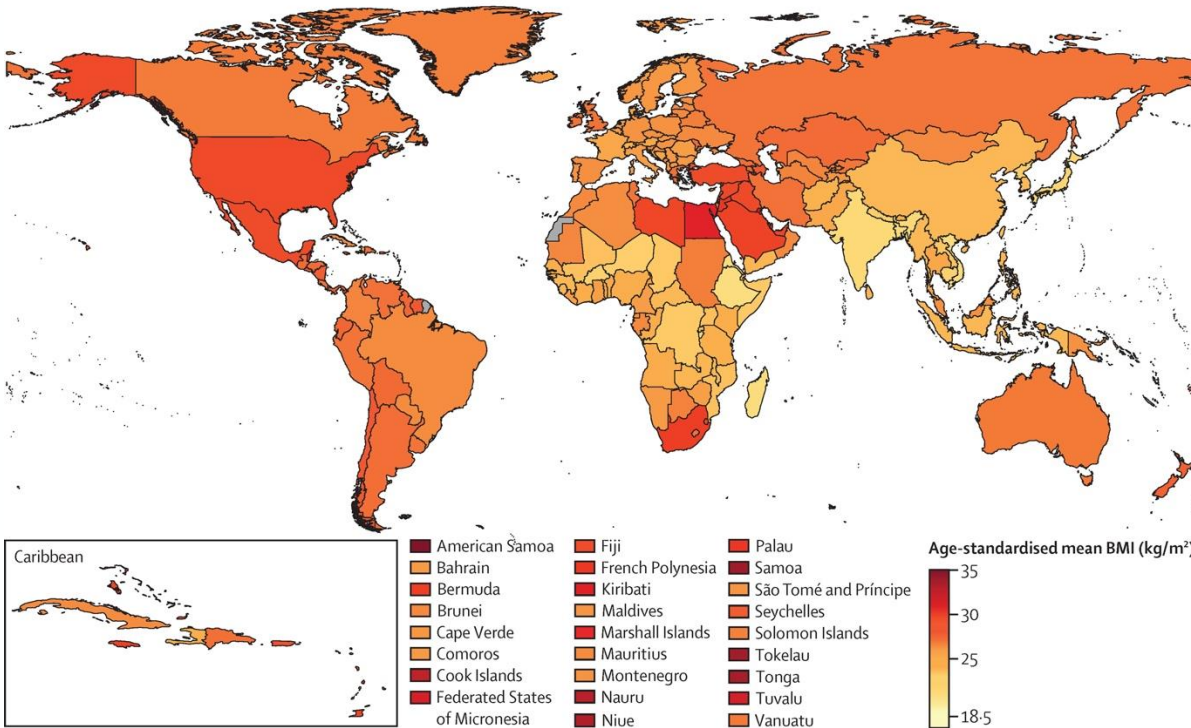
## 2. a) Differences among populations

## b) Clinal patterns (close together = similar)

### 1. Males vs females

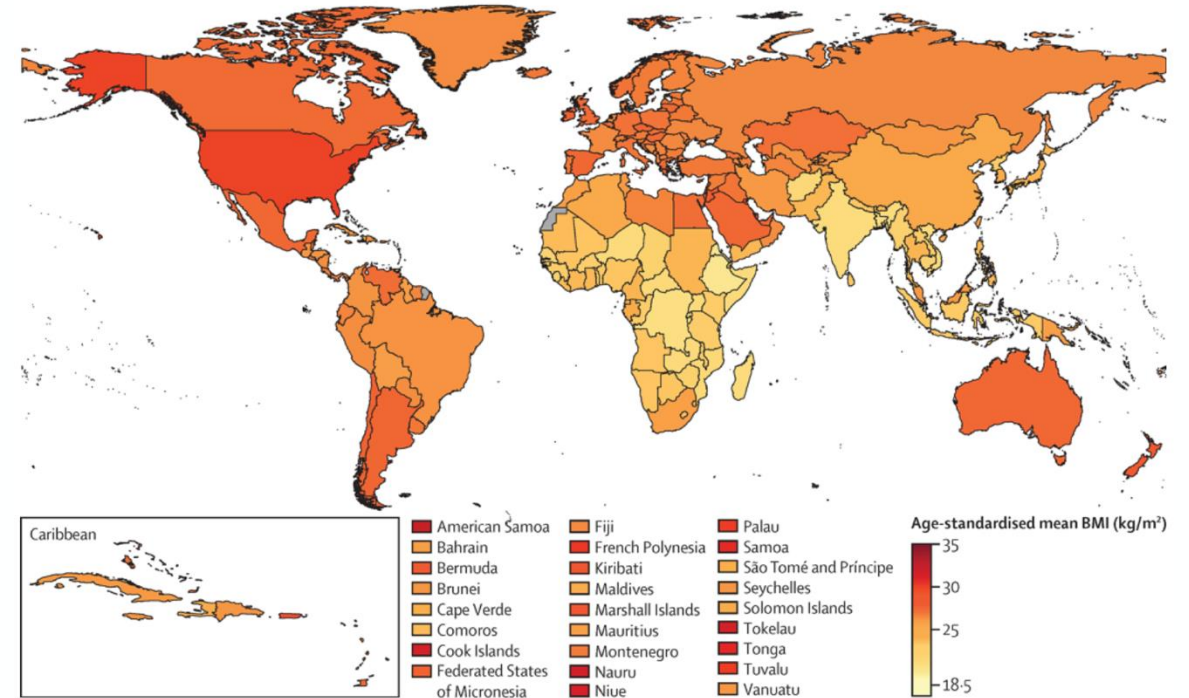
Female:

2014



Male:

2014



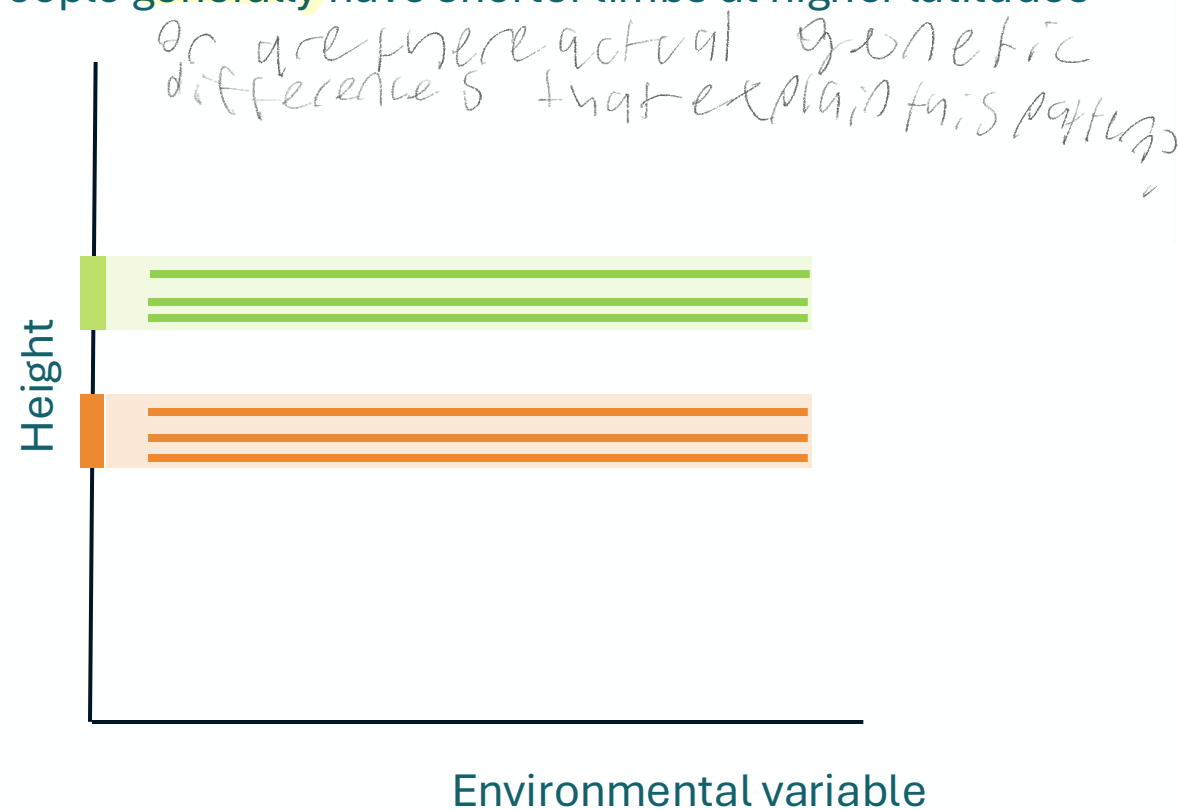
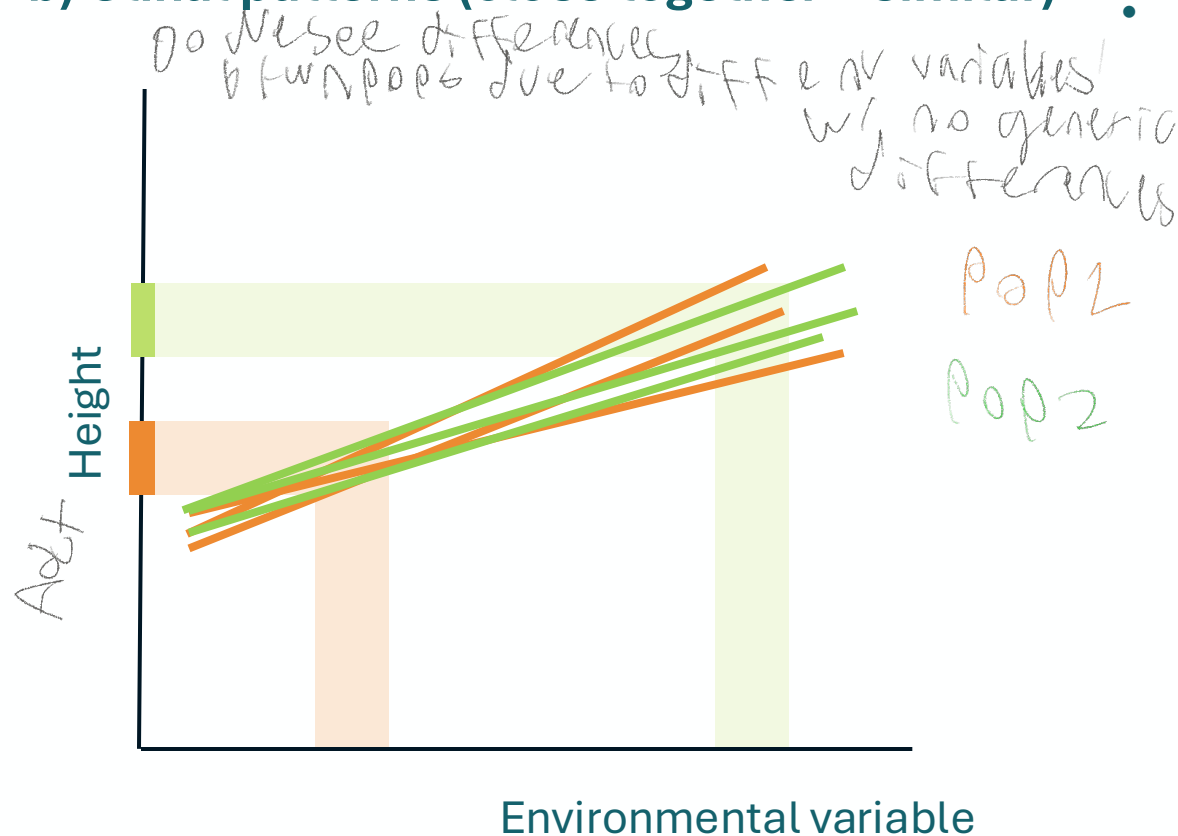
# General patterns: summary across all variables

## 1. Males vs females

## 2. a) Differences among populations

### b) Clinal patterns (close together = similar)

- People are **generally** taller at higher latitudes
- People **generally** have more fat at higher latitudes
- People **generally** have shorter limbs at higher latitudes



# General patterns: summary across all variables

## 1. Males vs females

## 2. a) Differences among populations

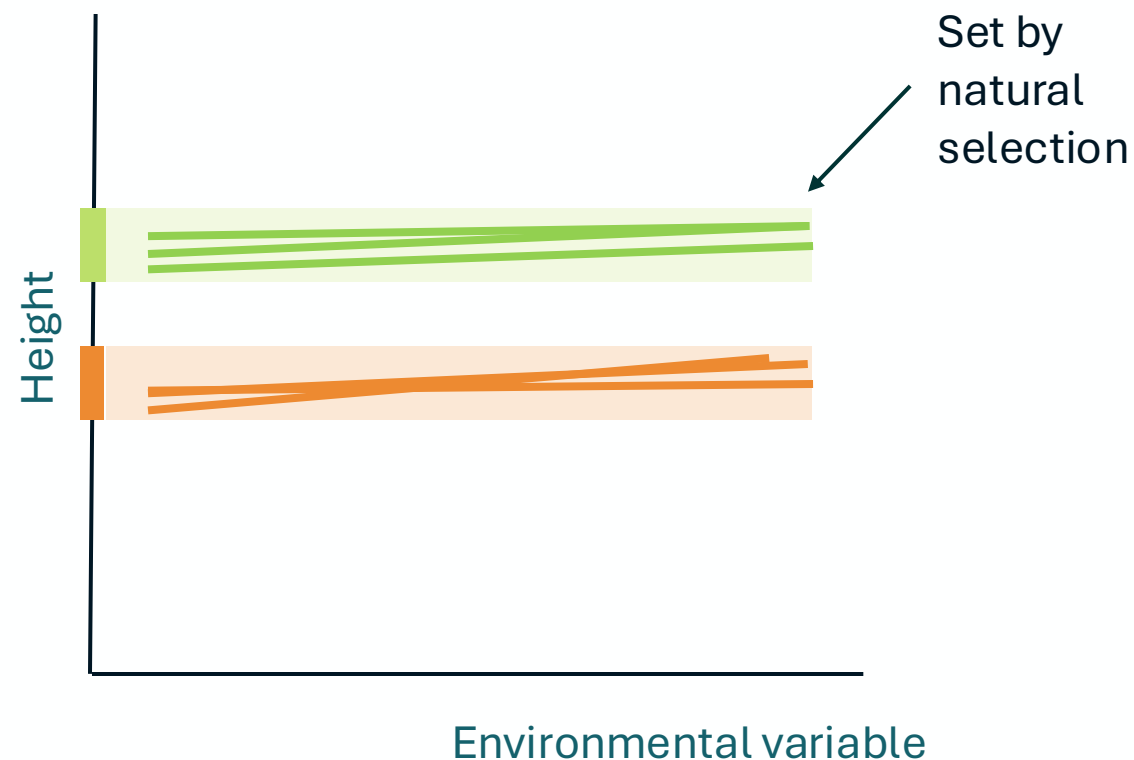
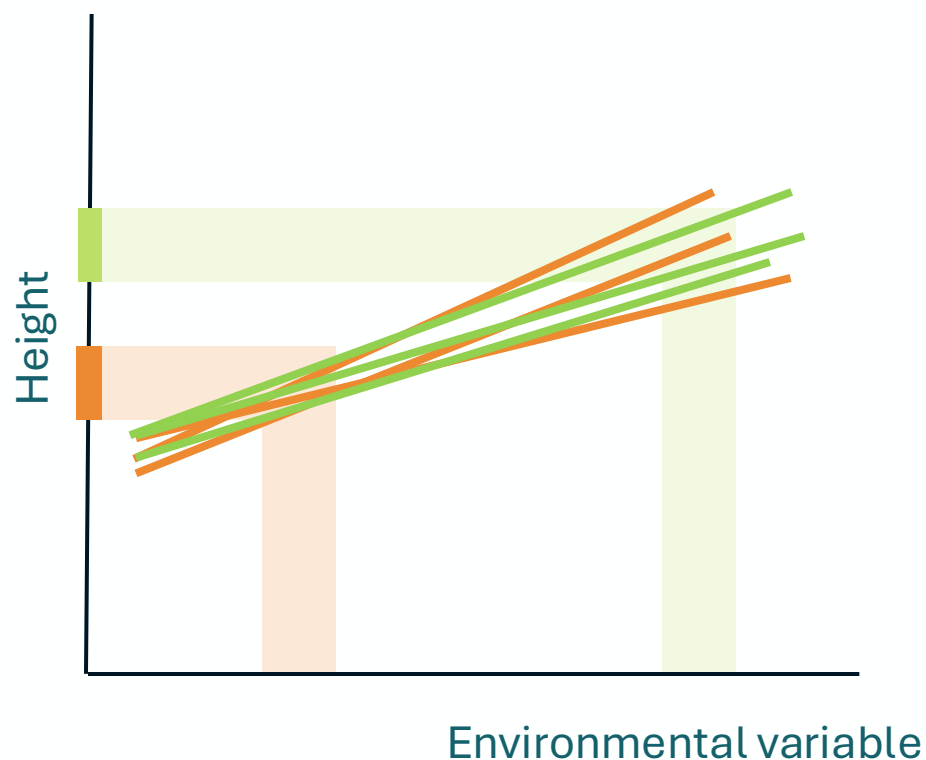
### b) Clinal patterns (close together = similar)

## 2 questions:

1. Which of these is correct?

2. If the first, what variable explains our pattern?

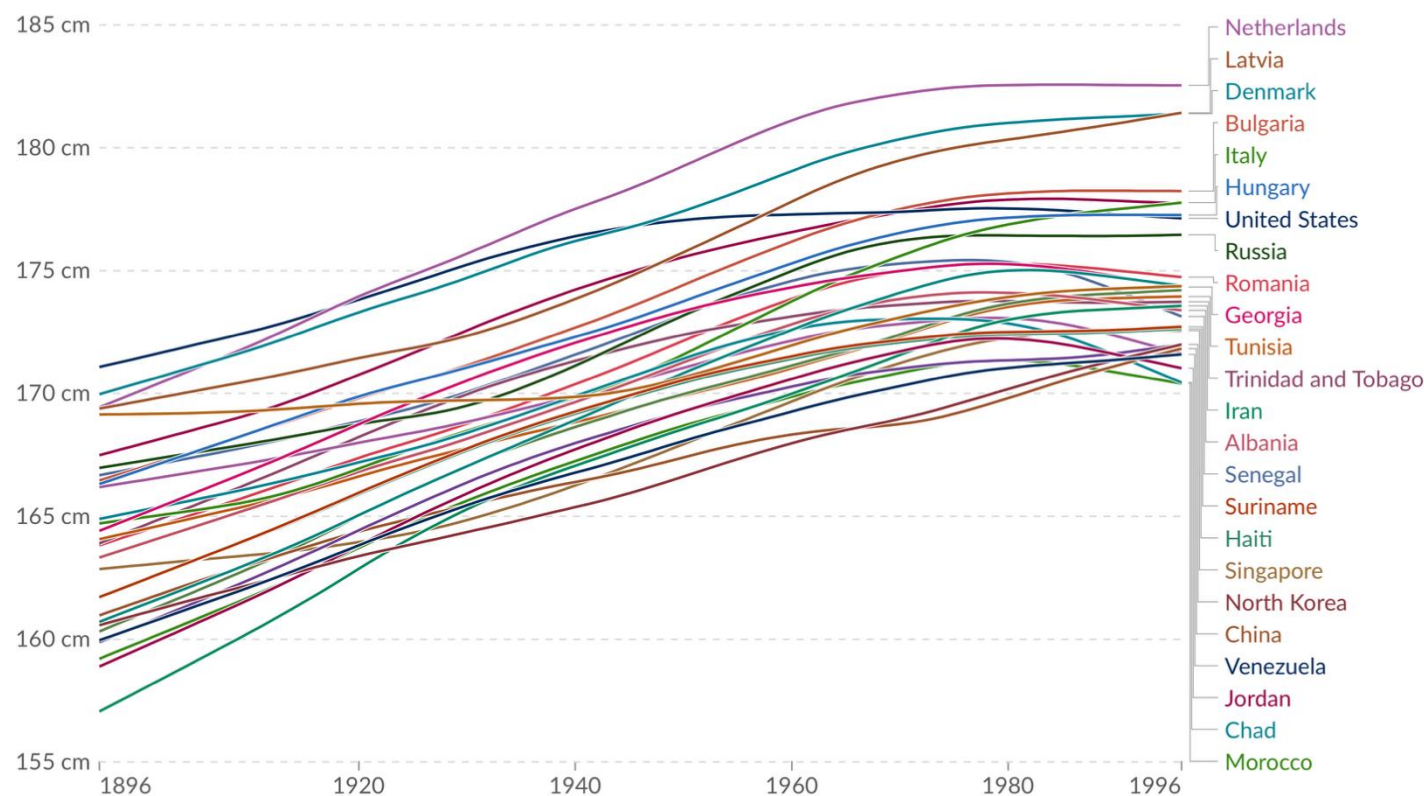
*dependence on environment or population?*



# Absolute genetic differences between populations?

## Average height of men by year of birth, 1896 to 1996

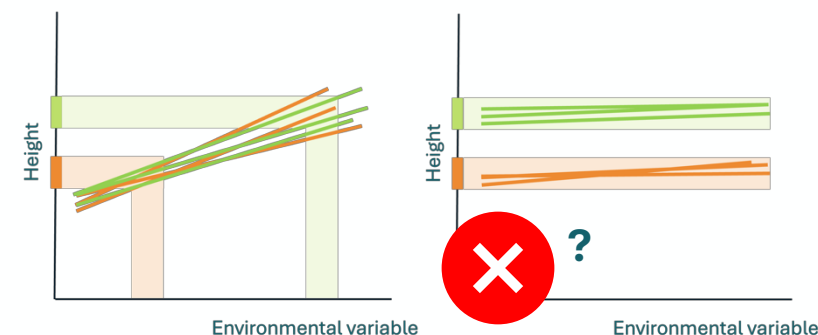
Mean height of adult men by year of birth. Data for the latest cohort (the year 1996) is therefore the mean height of men aged 18 in 2014.



Data source: NCD RisC, Human Height (2017)

OurWorldinData.org/human-height | CC BY

Our World  
in Data



Expect no  
variation over  
(recent) time

**unless...**



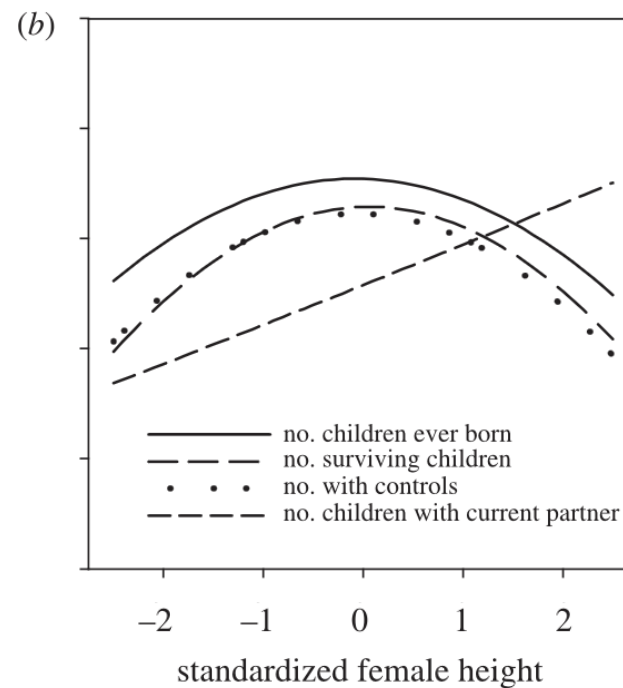
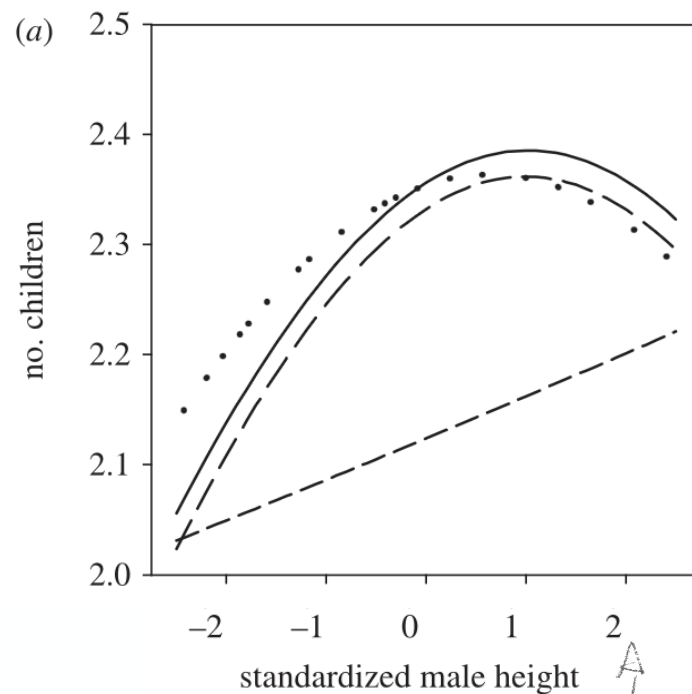
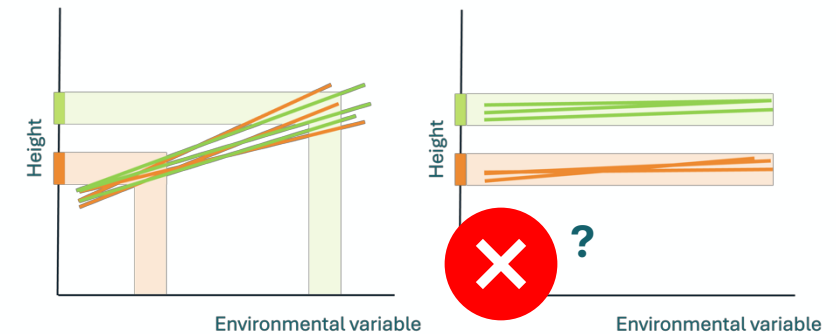
# Absolute genetic differences between populations?

## PROCEEDINGS B

rsb.royalsocietypublishing.org

### Does natural selection favour taller stature among the tallest people on earth?

Gert Stulp<sup>1,2</sup>, Louise Barrett<sup>3,4</sup>, Felix C. Tropf<sup>2</sup> and Melinda Mills<sup>5</sup>



correcting for confounders  
to find the  
people

Expect no  
variation over  
(recent) time

**unless...**

**but:**

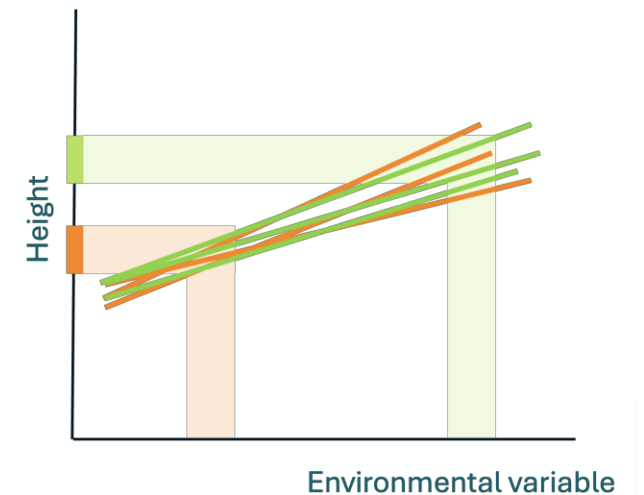
probably not due to natural selection

- Very small effect sizes
- Likely that this acts on other populations, too!

taller men → more kids ... don't look at genetic signals of selection

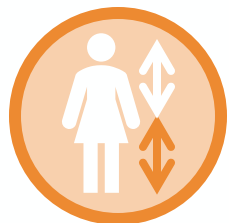
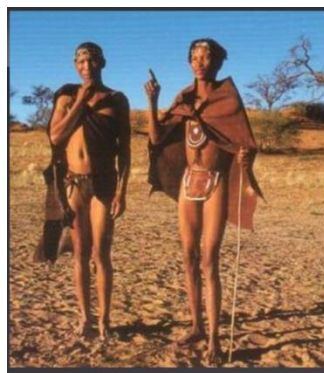
## 2. What variable(s) explain(s) our patterns?

- People are generally taller at higher latitudes
- People generally have more fat at higher latitudes
- People generally have shorter limbs at higher latitudes

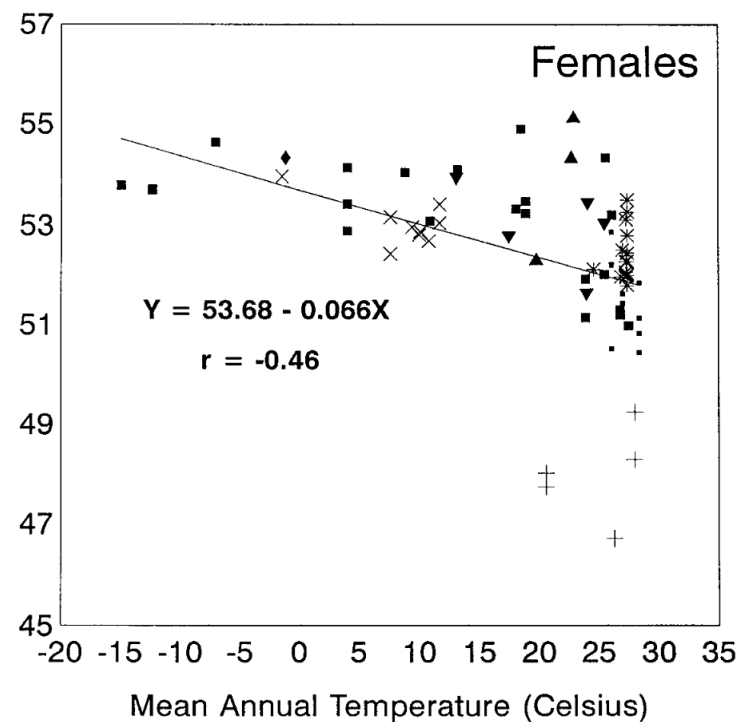


# Temperature

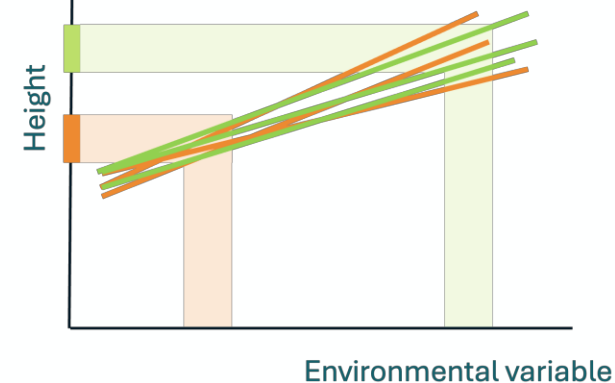
Looking  
at SHR



Longer limbs

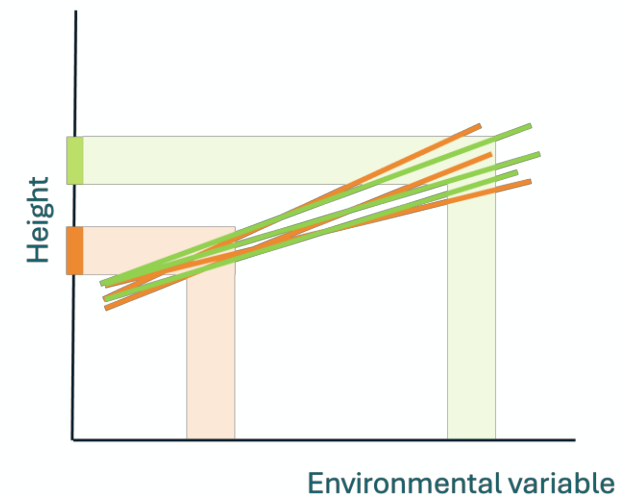
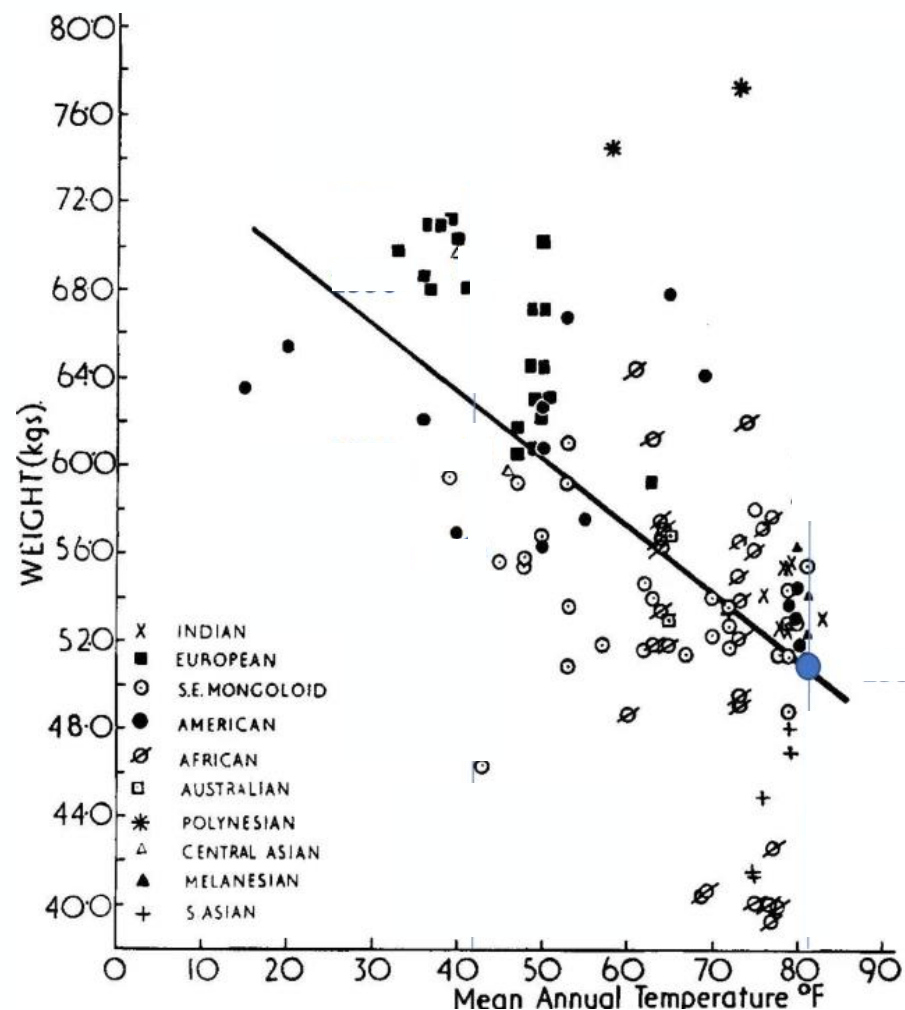
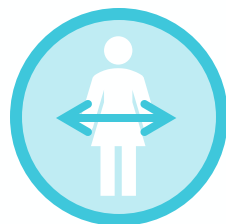


Review: lectures 3 & 4



Decreasing  
sitting height ratio  
↳ longer limbs

# Temperature

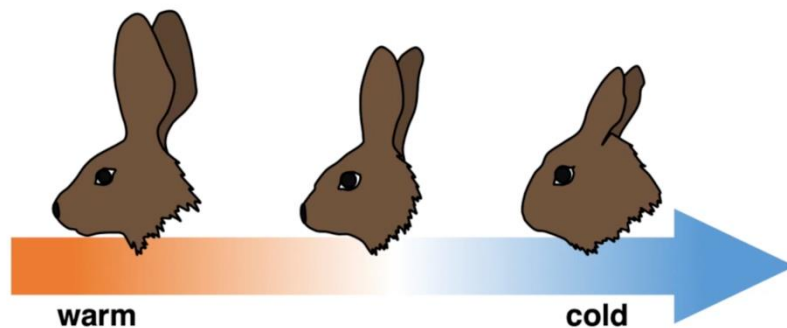
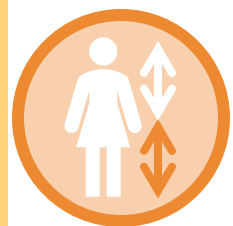




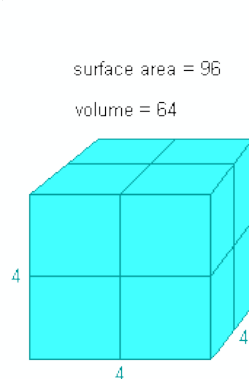
# Temperature

Review: lectures 3 &amp; 4

**Allen's rule** (Allen; limb length!):

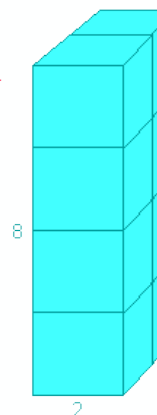


*What's supposed to be selected for is no bushes or trees to walk trees*

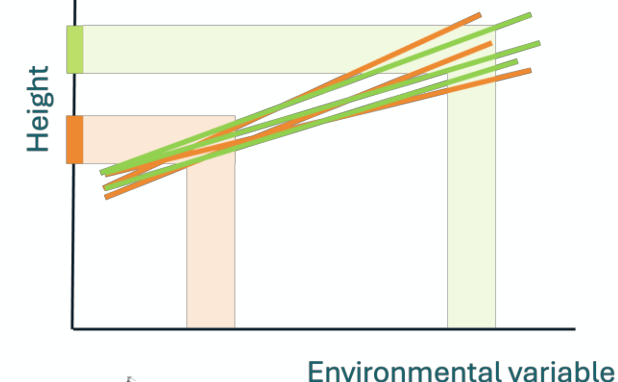


SA/volume ratio: small  
Heat dissipates slowly

surface area = 112  
1.75 times greater surface area  
volume = 64  
same volume

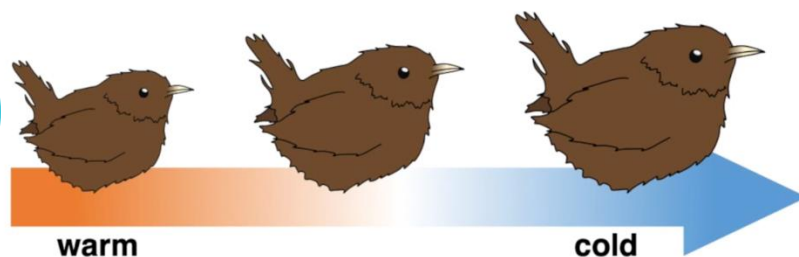
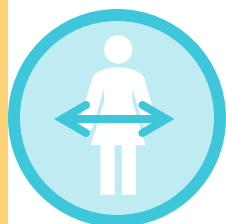


SA/volume ratio: big  
Heat dissipates fast

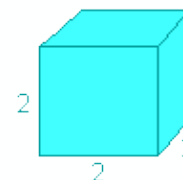


*Due to heat stress?*

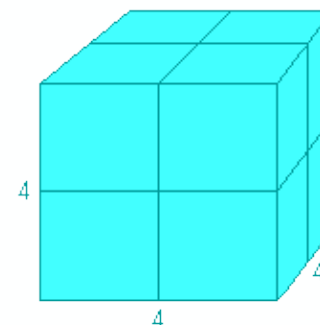
**Bergmann's rule** (Bergmann; body size):



surface area = 24  
(2 X 2)(6 sides)  
volume = 8  
(2 X 2 X 2)

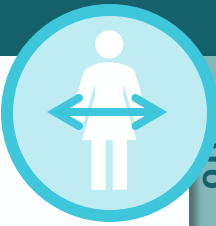


SA/volume ratio: big  
Heat dissipates fast



surface area = 96  
(4 X 4)(6 sides)  
4 times larger surface area  
volume = 64  
(4 X 4 X 4)  
8 times larger volume

SA/volume ratio: small  
Heat dissipates slowly

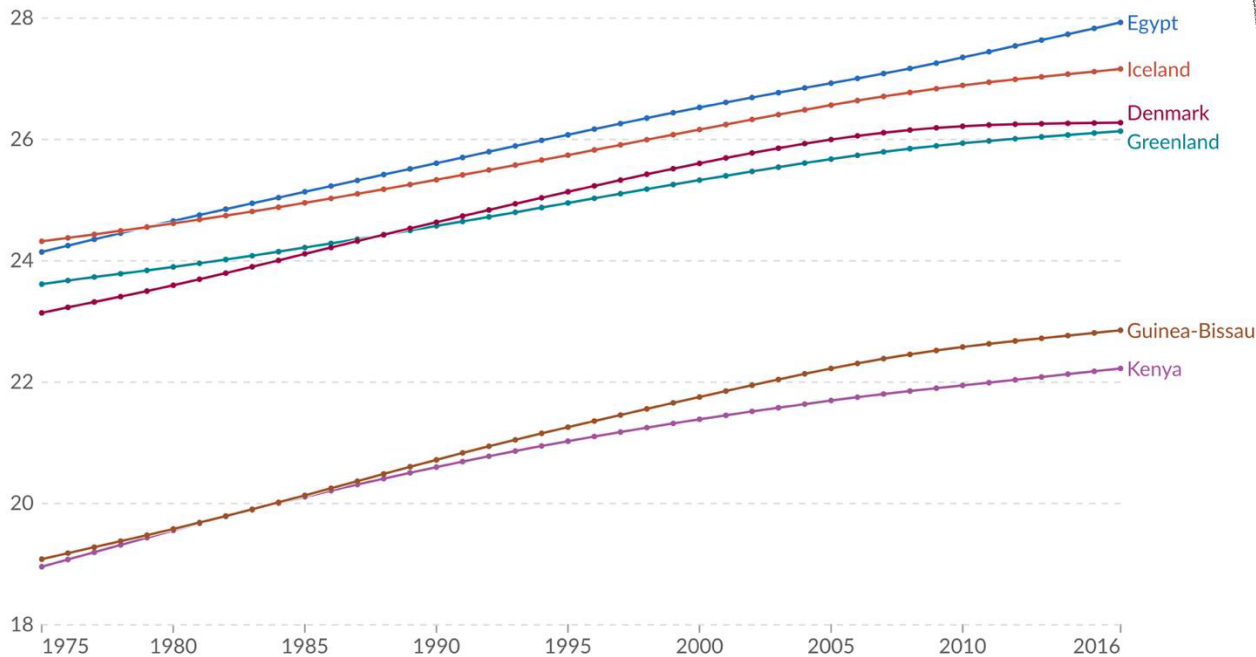


# Wait a minute...

## Mean body mass index (BMI) in men, 1975 to 2016

Our World in Data

Body mass index (BMI) is measured as a person's weight in kilograms (kg) divided by his height (in meters), squared. The WHO define a BMI  $\leq 18.5$  as 'underweight'; 18.5 to  $< 25$  as 'normal/healthy'; 25.0 to  $< 30$  as 'overweight'; and  $> 30.0$  as 'obese'.



Data source: NCD Risk Factor Collaboration (NCD-RisC)

OurWorldinData.org/obesity | CC BY

Note: To allow for comparisons between countries and over time, this metric is age-standardized<sup>1</sup>.

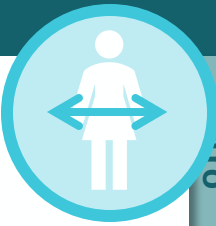
1. Age standardization Age standardization is an adjustment that makes it possible to compare populations with different age structures, by standardizing them to a common reference population.

Read more: [How does age standardization make health metrics comparable?](#)

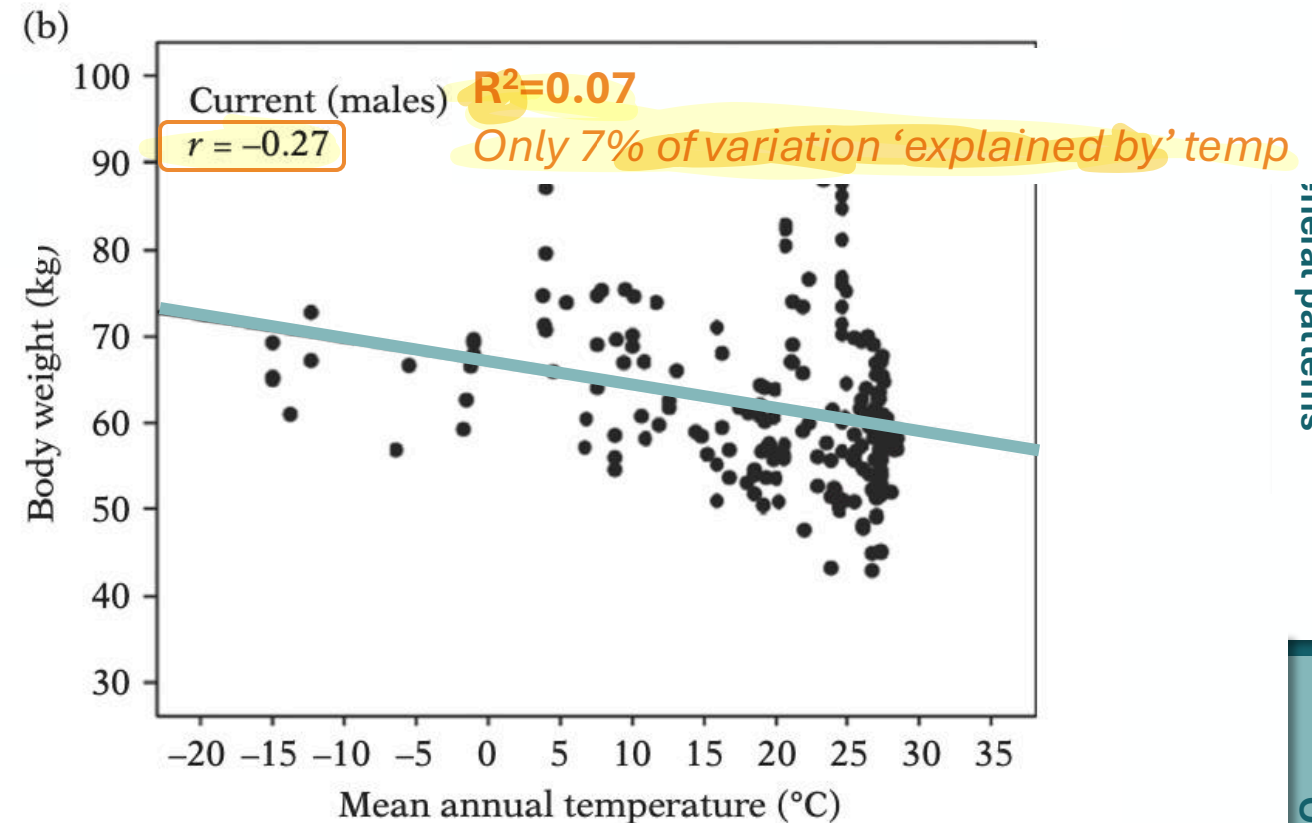
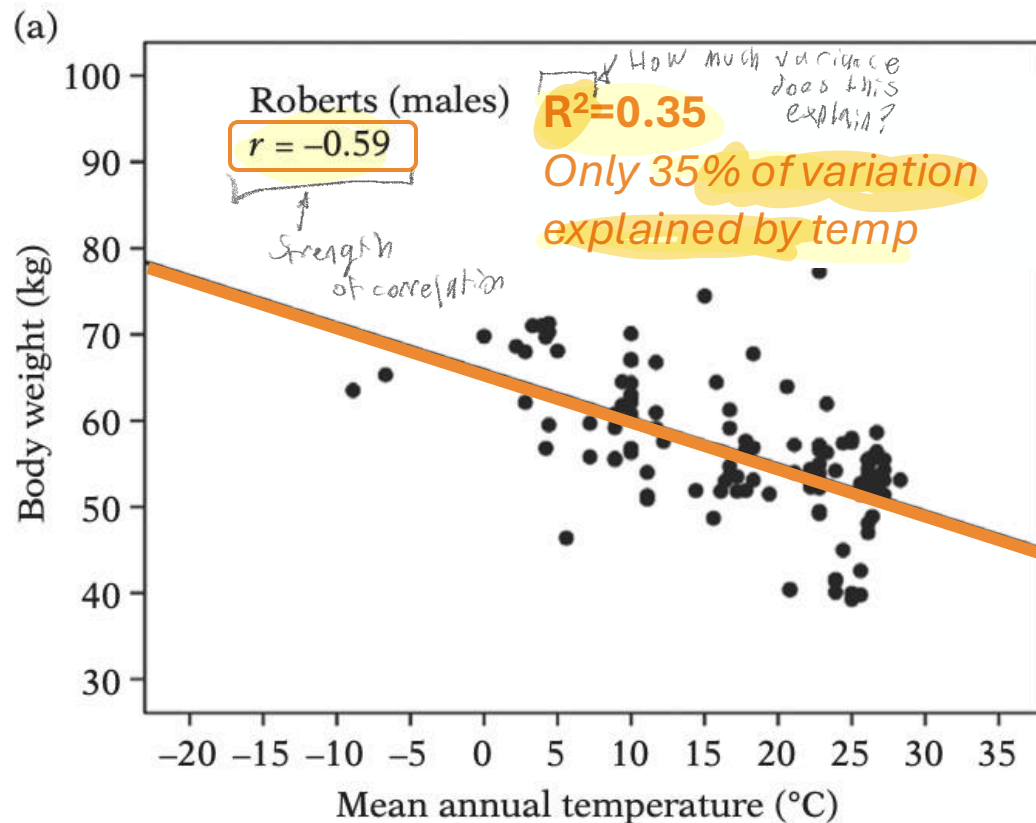
Increase in average BMI for everyone over time

## Problems with Bergmann's Rule

- Temperature has not changed (much-except for going up!) you'd expect to see the opposite
- Increase in sub-Saharan African populations not expected under Bergmann's Rule
- Pattern for Egypt *definitely* not expected under Bergmann's Rule

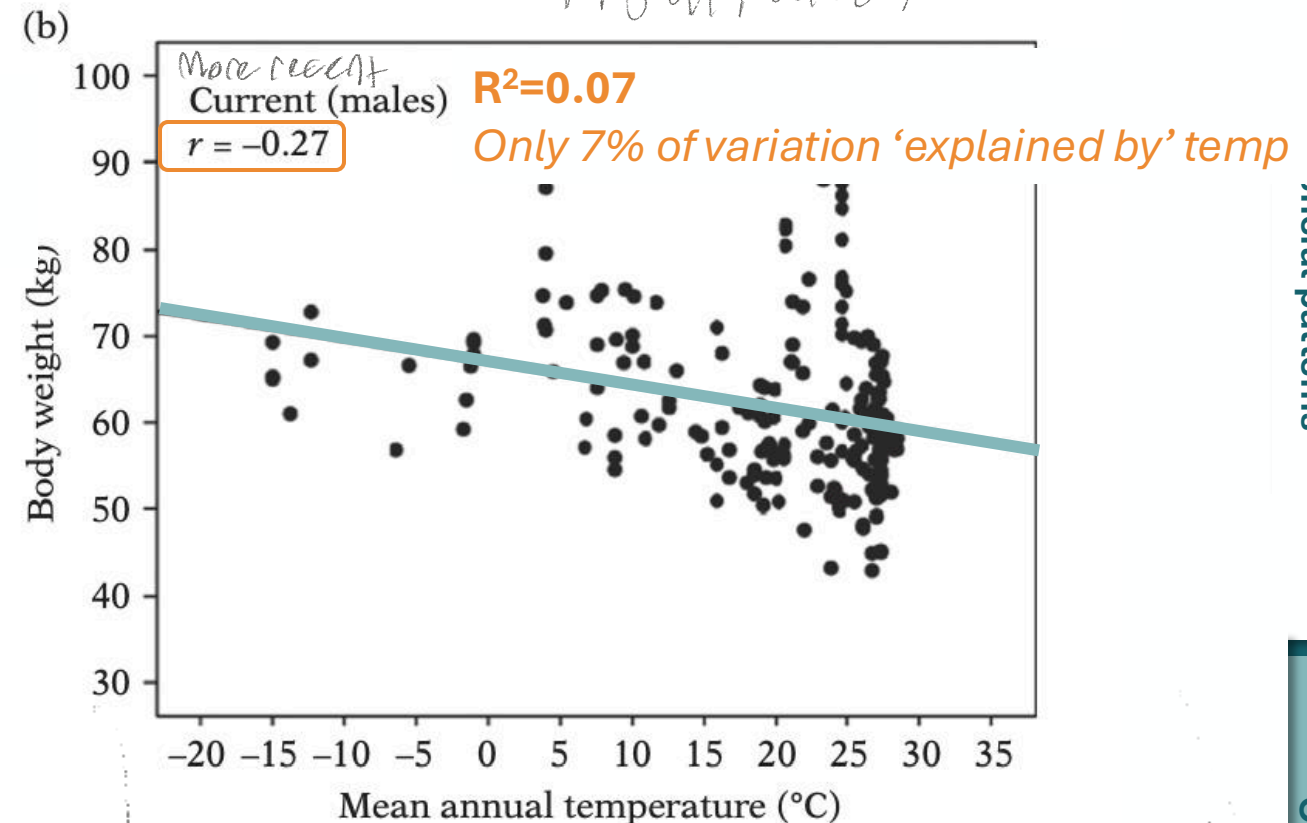
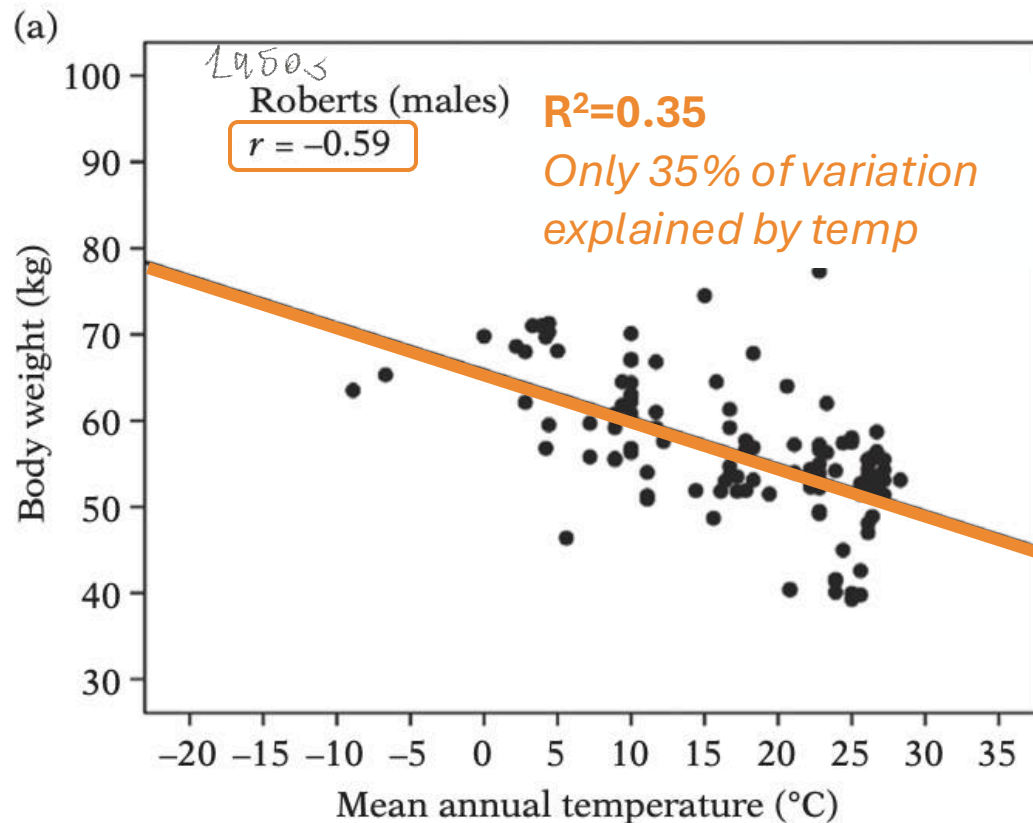


# Wait a minute...





# New problems: why did Roberts find a pattern, and why does it disappear in 40 years?



Relationship almost regularly  
disappears?

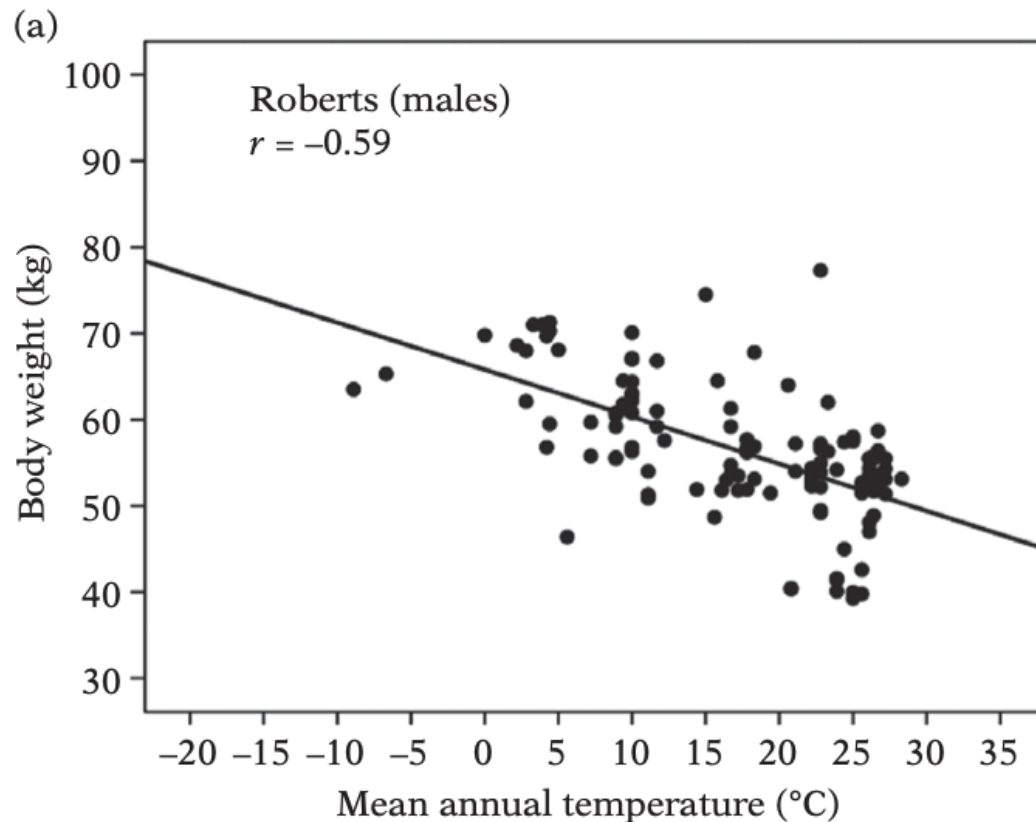
more recent data shows  
weaker correlation  
weight & temp





# New problem: why did Roberts find a pattern?

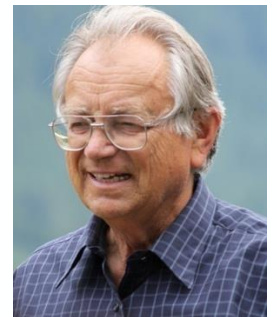
What is actually causing this is nutrients & not a response to heat stress



“...[Bergmann’s association] ... follows the duration of the annual productivity pulse, so that **body size is a function of availability of nutrients and energy during periods of growth.**

Correlations between body size and temperature are shown to be spurious. If reduction in relative surface area is indeed an adaptation to conserve heat, then mammals should increase in size from south to north at rates two orders of magnitude greater than they do.”

Valerius Geist

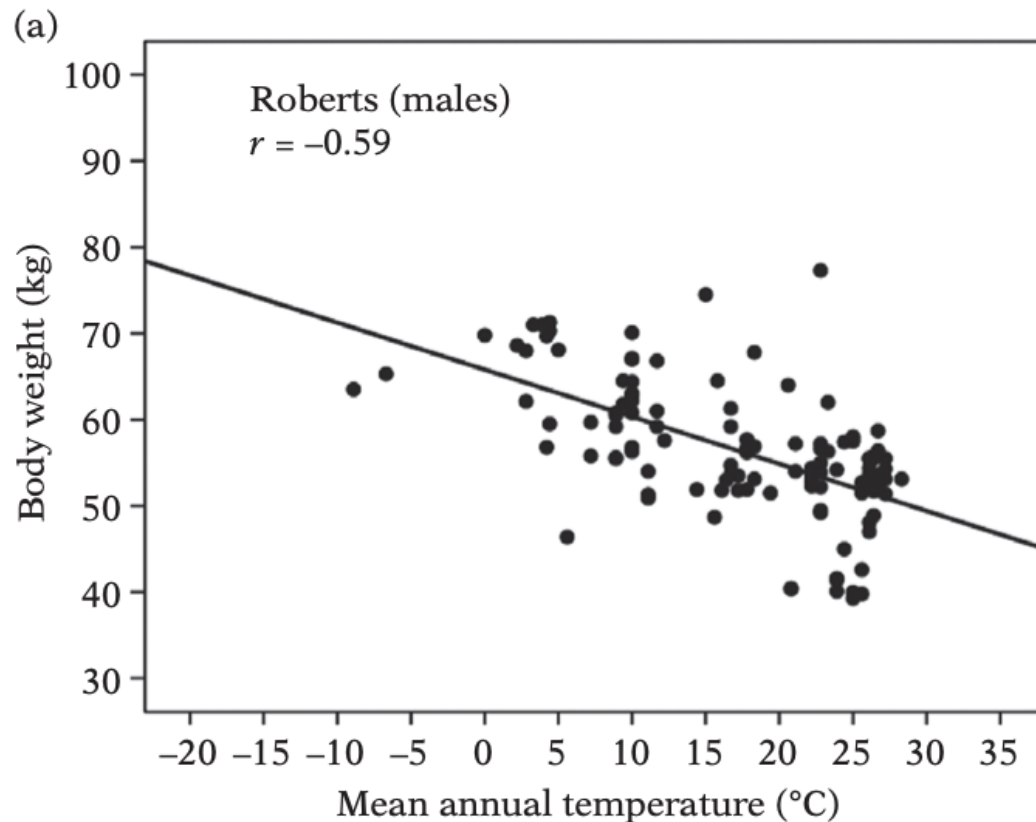


Geist. *Can J. Zool.* (1987)



# New problem: why did Roberts find a pattern?

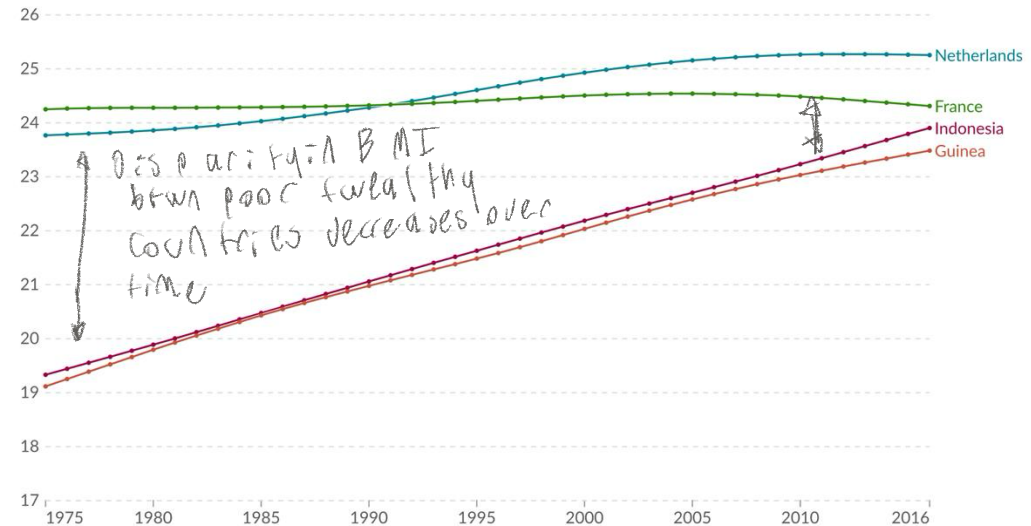
Roberts was really describing wealth and health disparities between WEIRD countries and LMICs in 1953. These persist today – but globalization has *enormously* changed nutrition.



## Mean body mass index (BMI) in women, 1975 to 2016

Body mass index (BMI) is measured as a person's weight in kilograms (kg) divided by her height (in meters), squared. The WHO define a BMI  $\leq 18.5$  as 'underweight'; 18.5 to  $< 25$  as 'normal/healthy'; 25.0 to  $< 30$  as 'overweight'; and  $> 30.0$  as 'obese'.

Our World in Data



Data source: NCD Risk Factor Collaboration (NCD-RisC)

OurWorldinData.org/obesity | CC BY

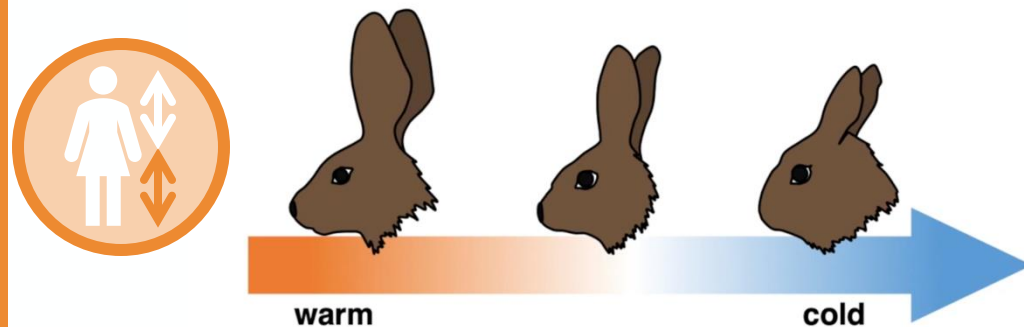
Note: To allow for comparisons between countries and over time, this metric is age-standardized<sup>1</sup>.

1. Age standardization Age standardization is an adjustment that makes it possible to compare populations with different age structures, by standardizing them to a common reference population.

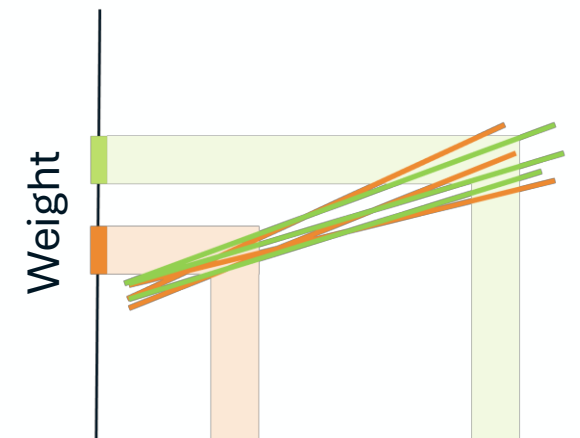
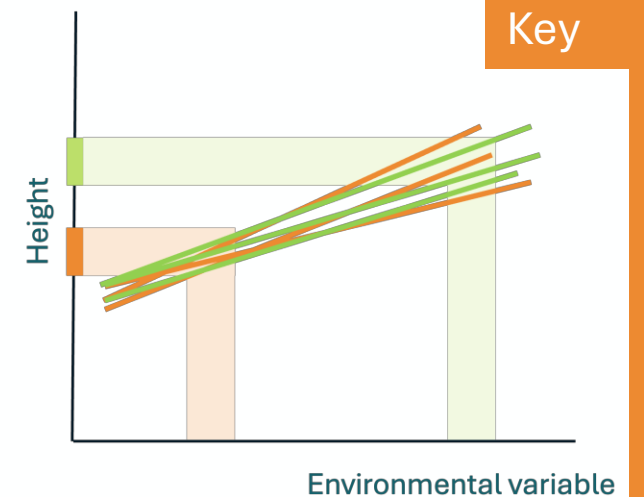
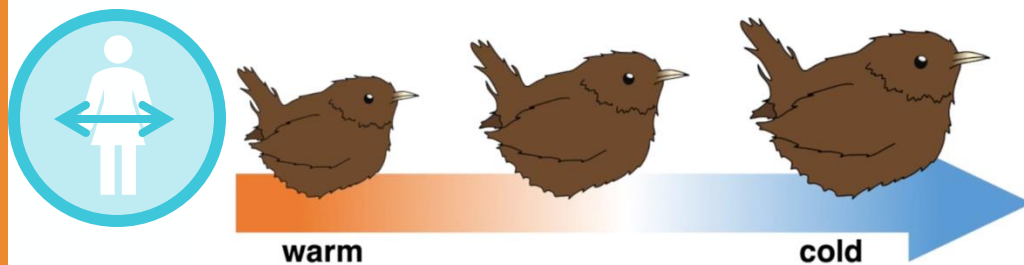
Read more: [How does age standardization make health metrics comparable?](#)

# Temperature

**Allen's rule** (Allen; limb length!):

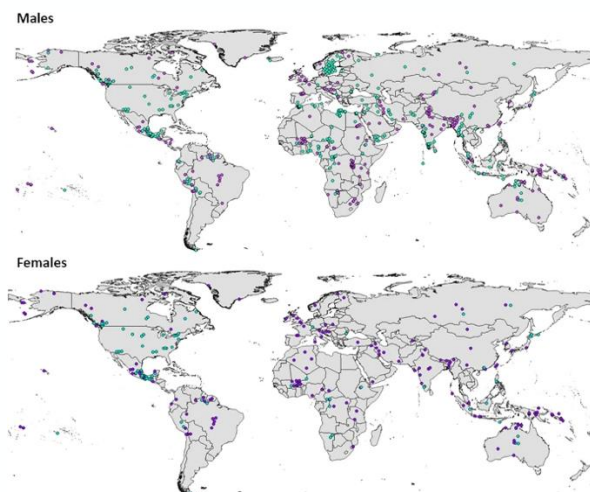


**Bergmann's rule** (Bergmann; body size):

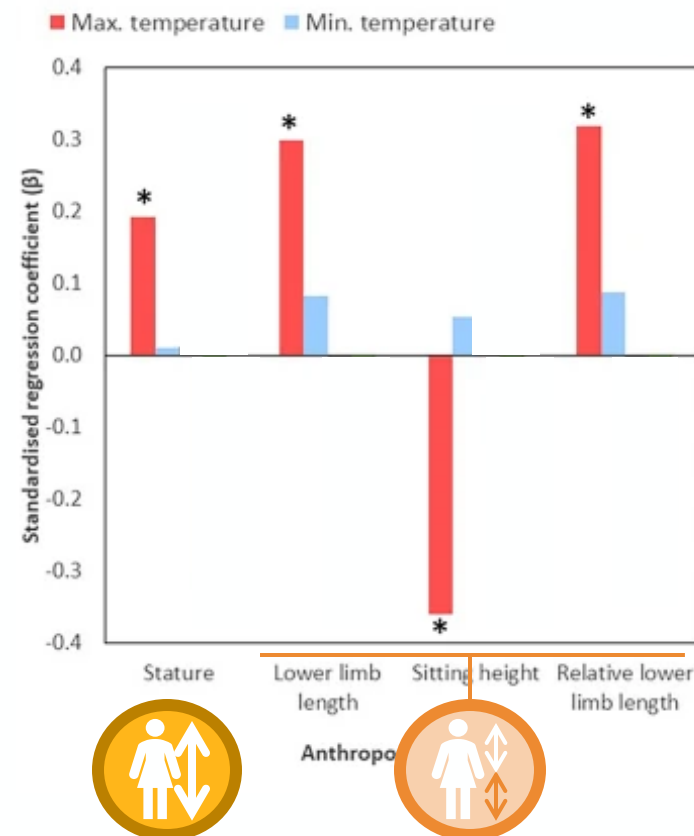


Not temperature  
Nutrition during growth is key

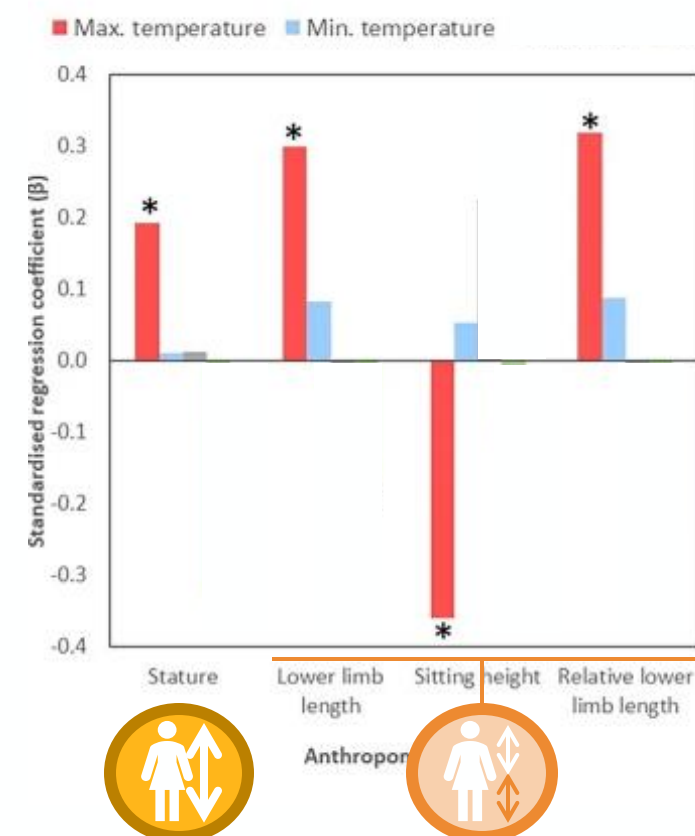
# Modern-day data provide (some) support for Allen's rule (and Bergmann's rule as applied to stature)



Males:

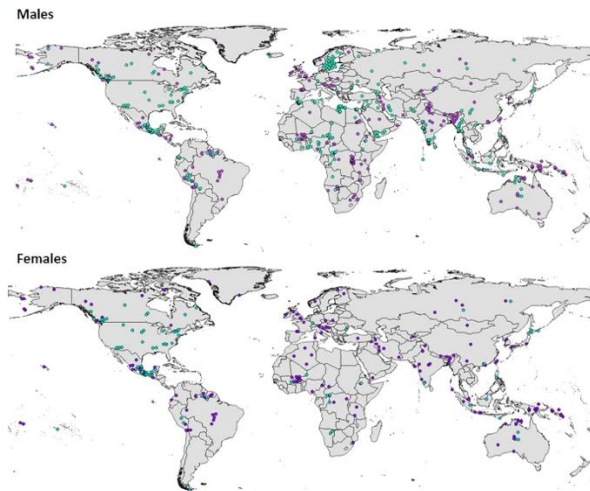


Females:

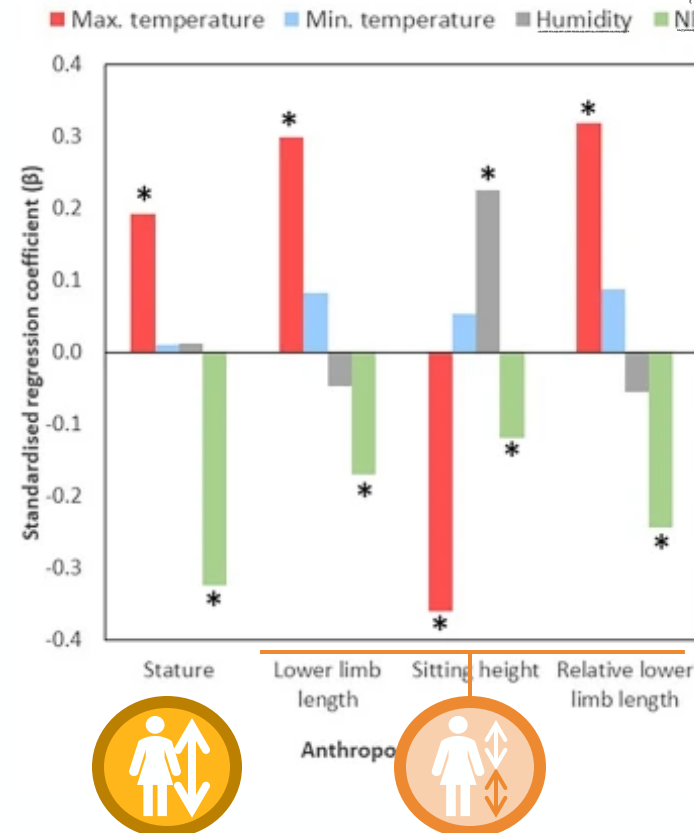




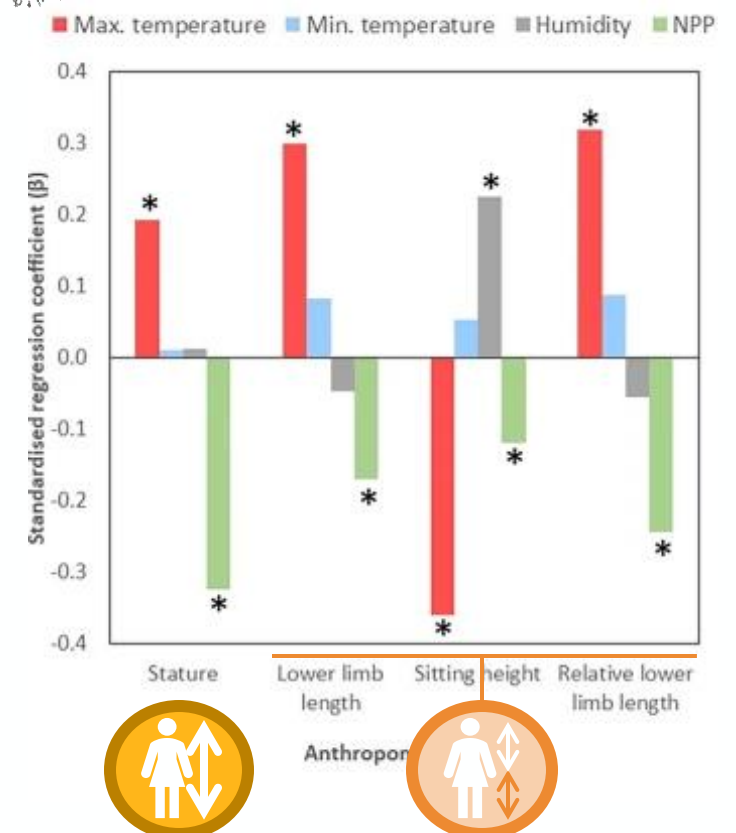
Modern-day data provide (some) support for Allen's rule (and Bergmann's rule as applied to stature) **but other factors likely play more important roles**



Males:

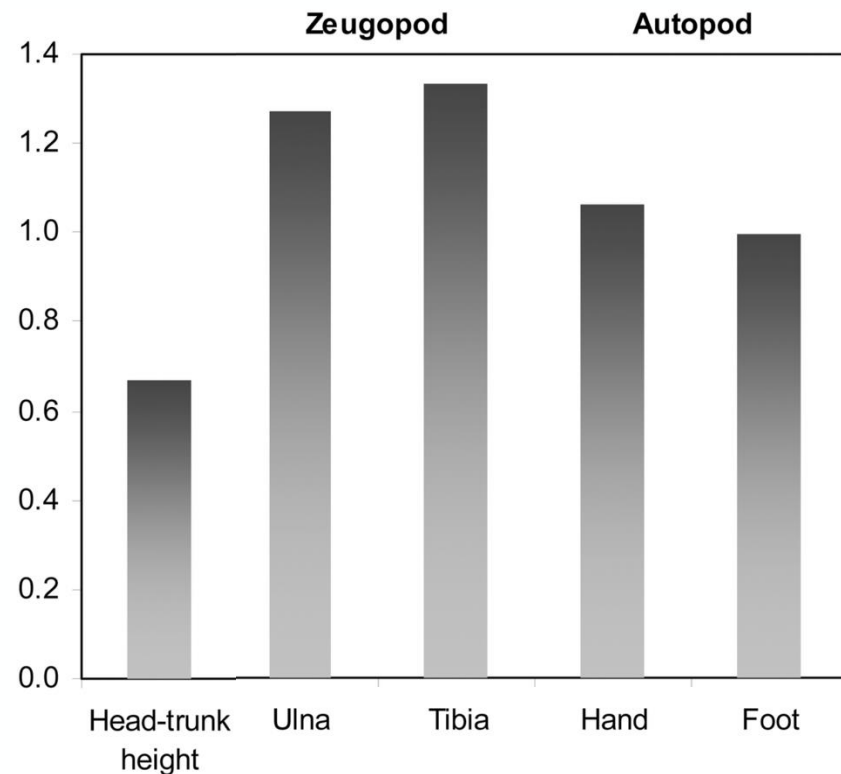


Females:



Modern-day data provide (some) support for Allen's rule (and Bergmann's rule as applied to stature) **but other factors likely play more important roles**

Difference between  
lowland (not stressed)  
and highland (stressed)



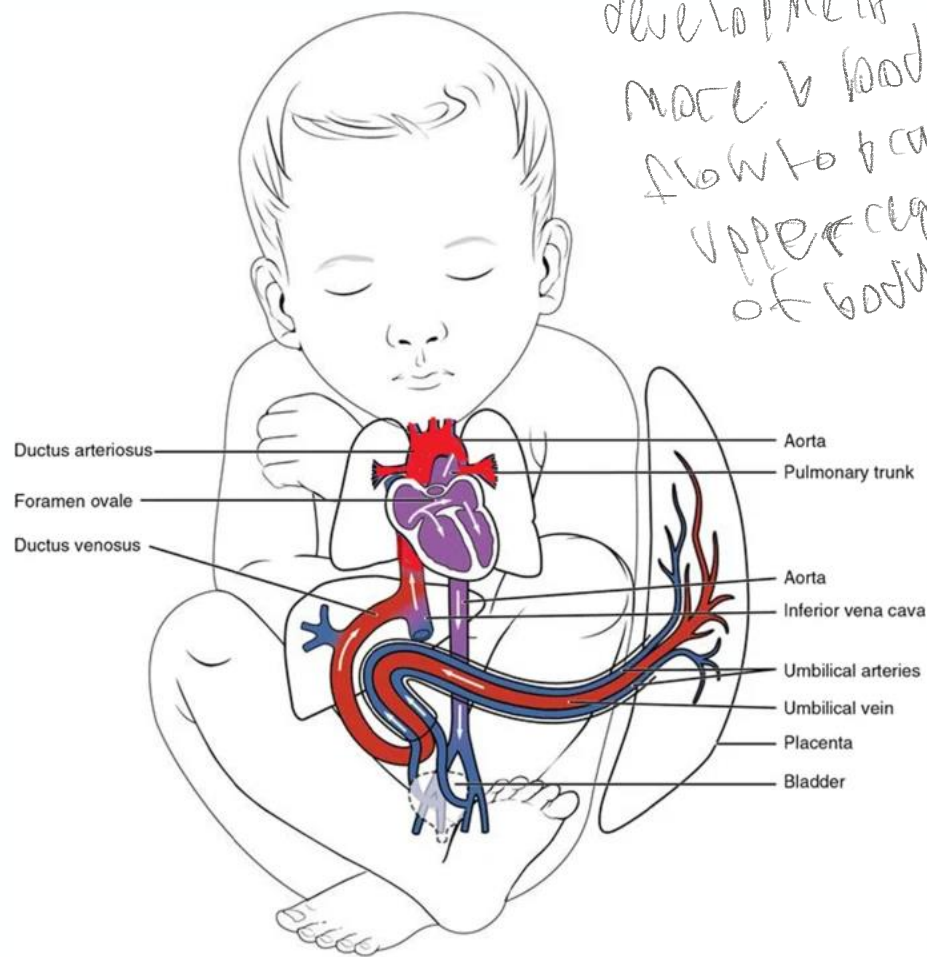
**Ulnae and tibiae are most sensitive to nutritional stress, and differences in their lengths contribute most to differences in stature and limb proportions between high- and lowland populations**



How fetal circulatory system operates

## Proximate:

during fetal development - more blood flow to brain + upper regions of body



Ulnae and tibiae are most sensitive to nutritional stress

## Ultimate:



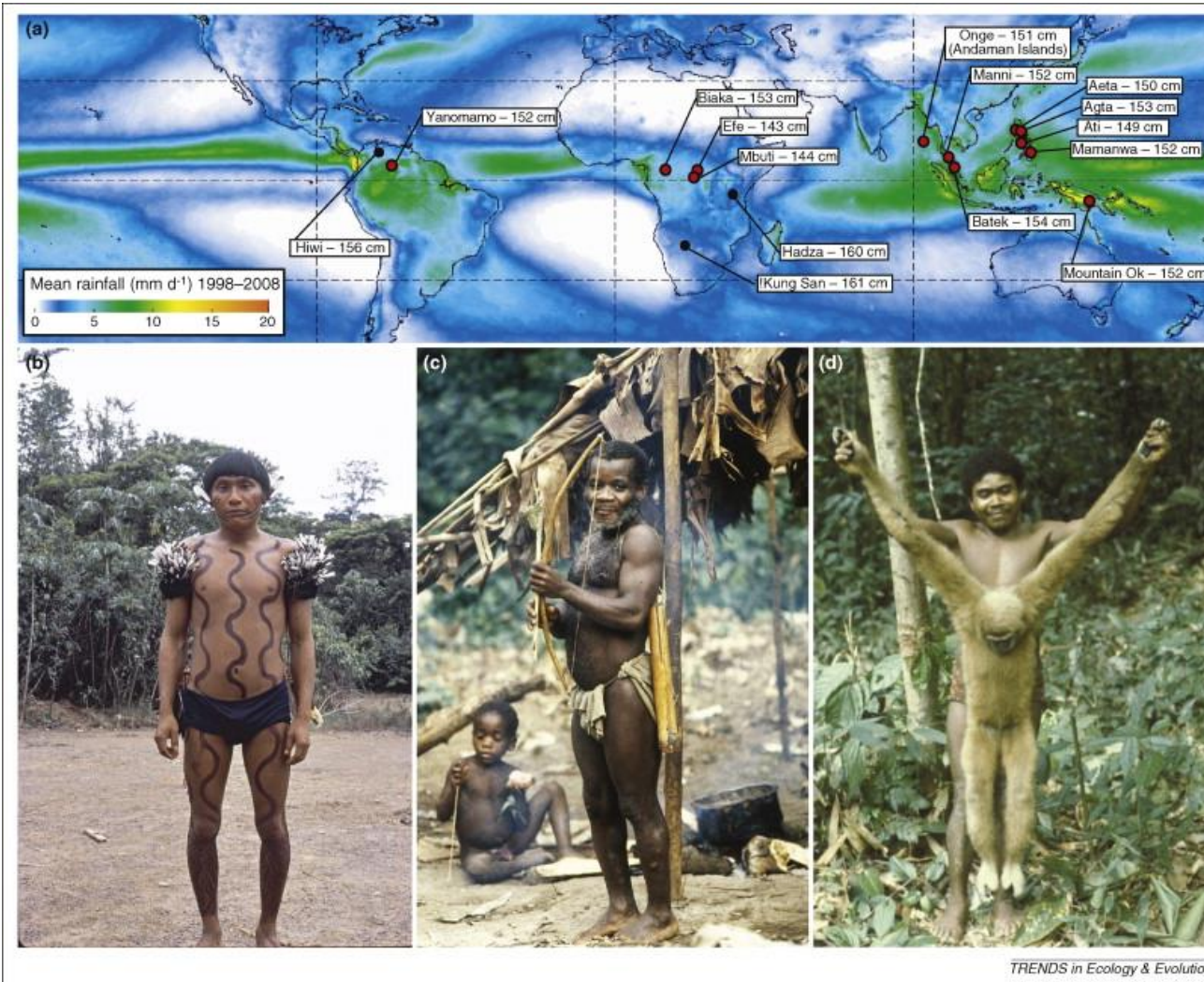
you can do without your legs but not your head - if you need to make tradeoffs

Tinbergen





# Interesting outliers to these general patterns



In groups, brainstorm potential drivers – whether proximate or ultimate! – of pygmy stature.

Choose one, and design a research question that you might use to inform a research project.

(Practice for your Research Project Design).

Proximate  
→ Immune responses  
vs growth

Ultimate  
is assumption  
that immune  
responses & growth  
are traded off  
→ tradeoffs outside of  
pick vs vit paradigm

→ disease rates + mortality  
rates vs height - how to differentiate  
between correlation & causation  
Track of growth hormones in diseased inds  
vs not disease  
→ Immune markers in blood to  
see if they are actively in immune  
response  
Activity of growth hormones vs  
activity of immune markers from  
same inds at same point in time  
→ go over multiple  
time points

# Summary of today's lecture

## General patterns

- Males tend to be taller, have longer limbs vs trunk, and be heavier than females
- People are generally taller at higher latitudes
- People generally have more fat at higher latitudes
- People generally have shorter limbs at higher latitudes
- Pygmy phenotypes

## General processes

Big role for genes, but lots of overlap between males and females, and a huge effect of environmental variables

Allen's Rule holds for limb lengths; recent data does *not* support Bergmann's rule.

In all cases, secular trends *and* specific studies show that nutrition during growth plays a major role.

Nutrition generally affects extremities most!

‘Caused’ by an early cessation of growth – not a decreased *rate* of growth. As for the ultimate reason... the jury is still out!